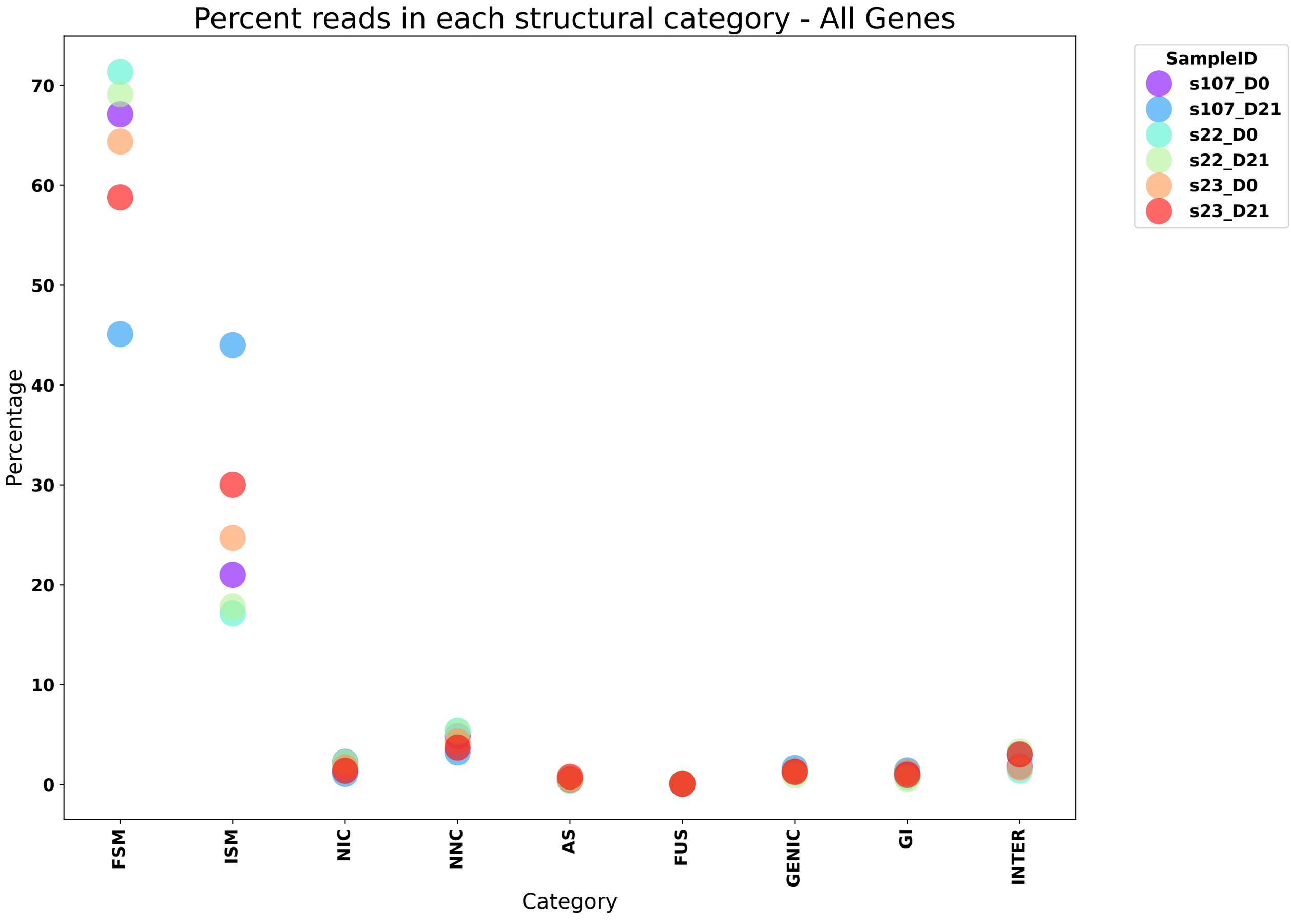
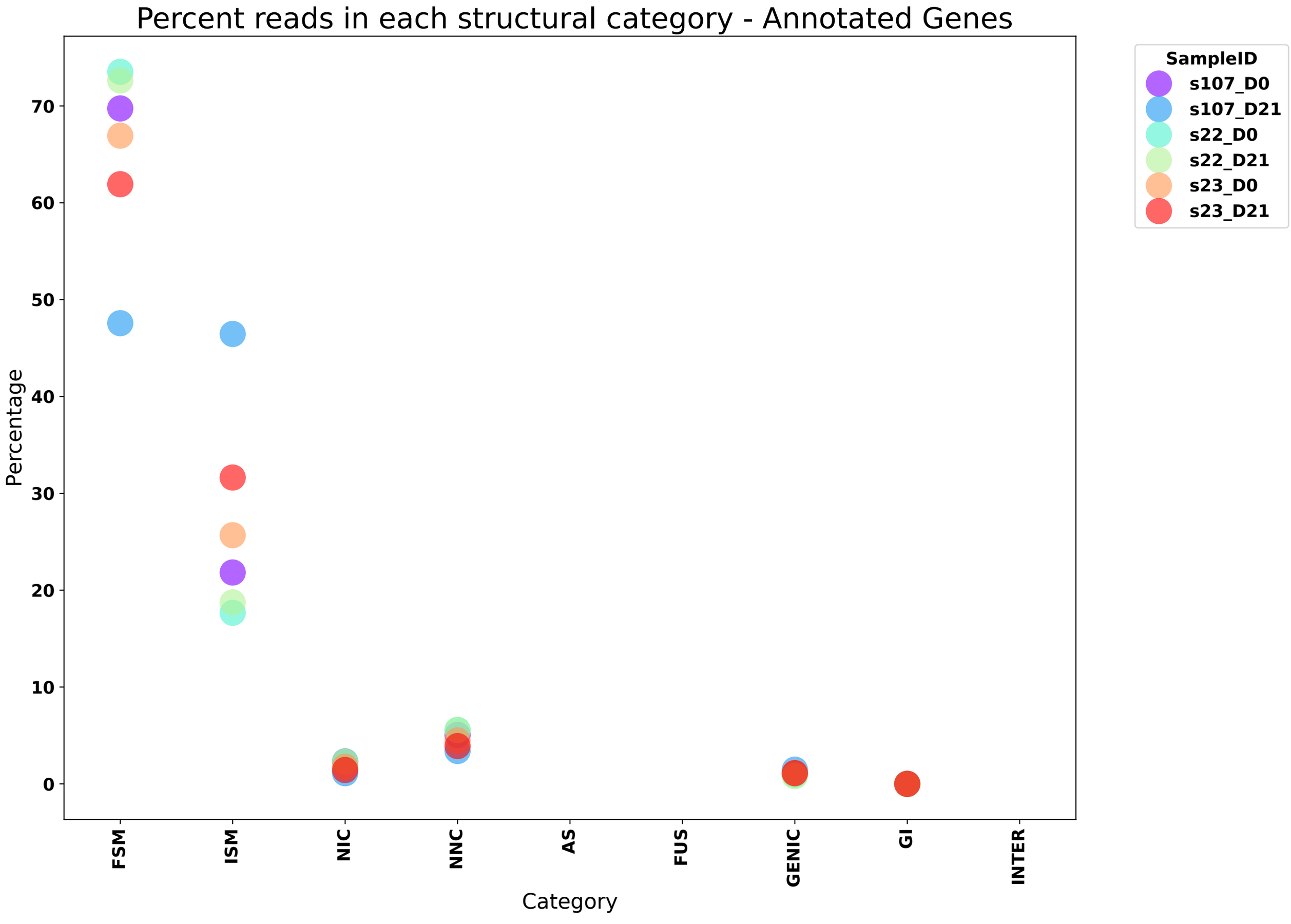
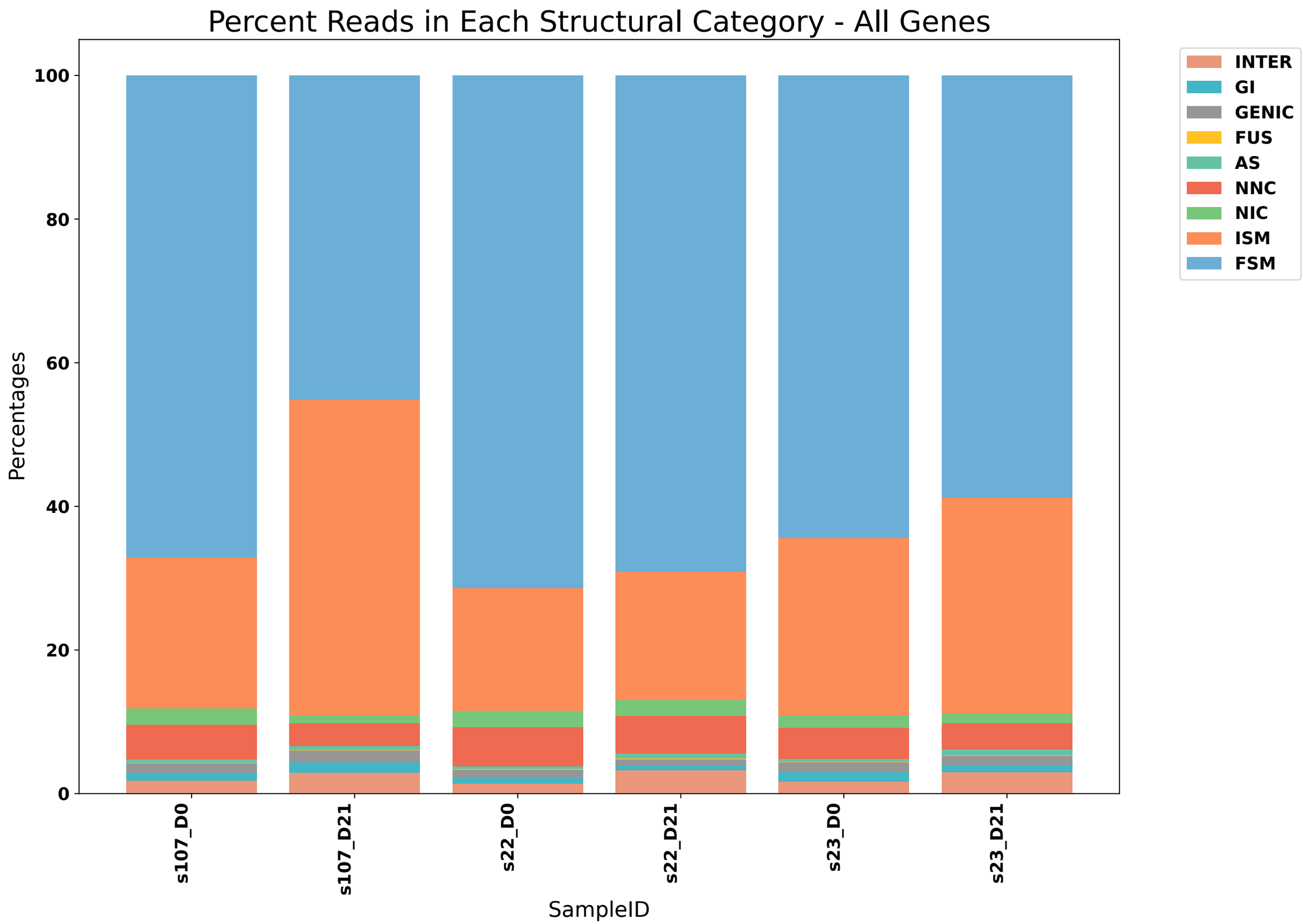
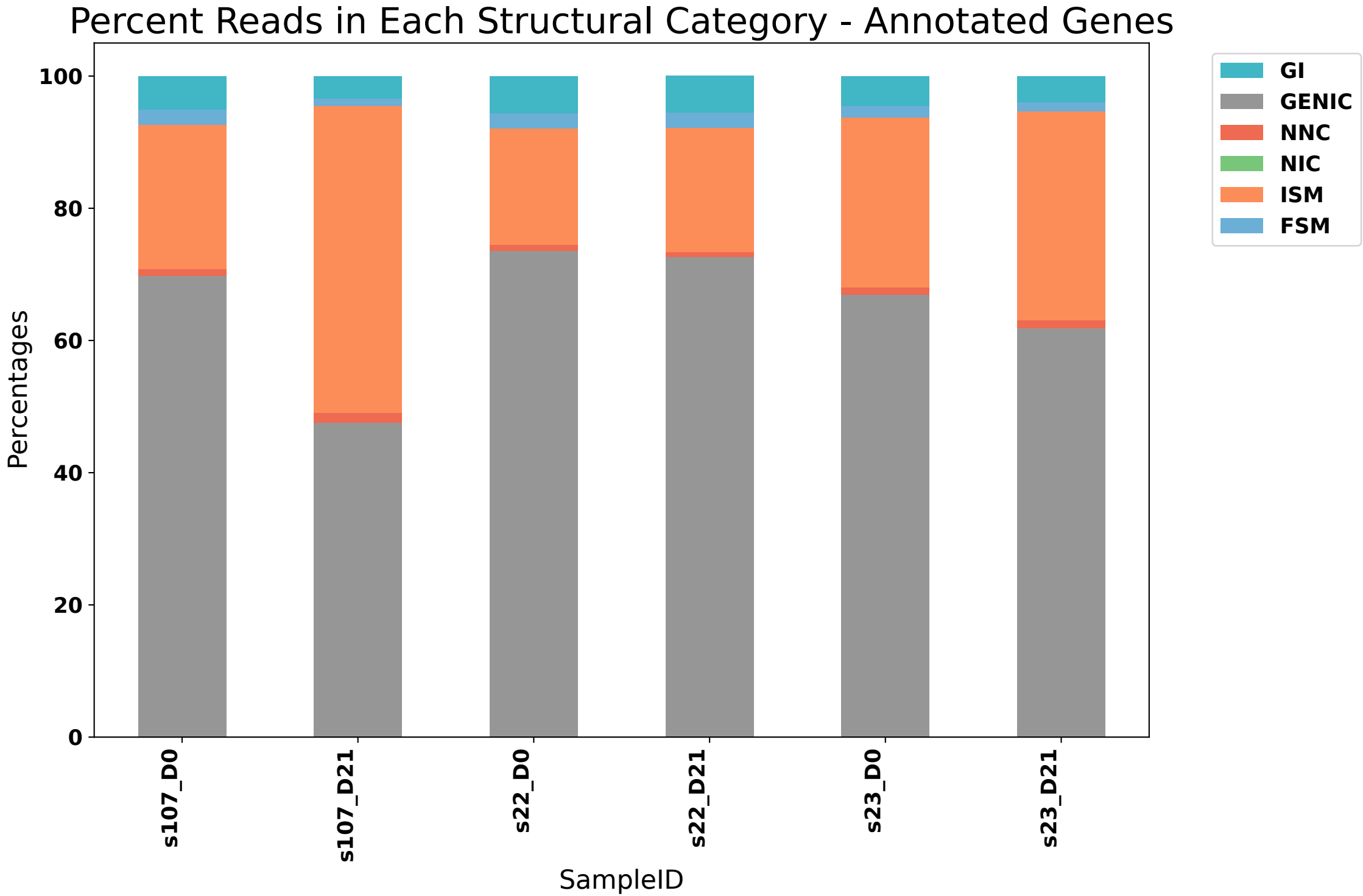
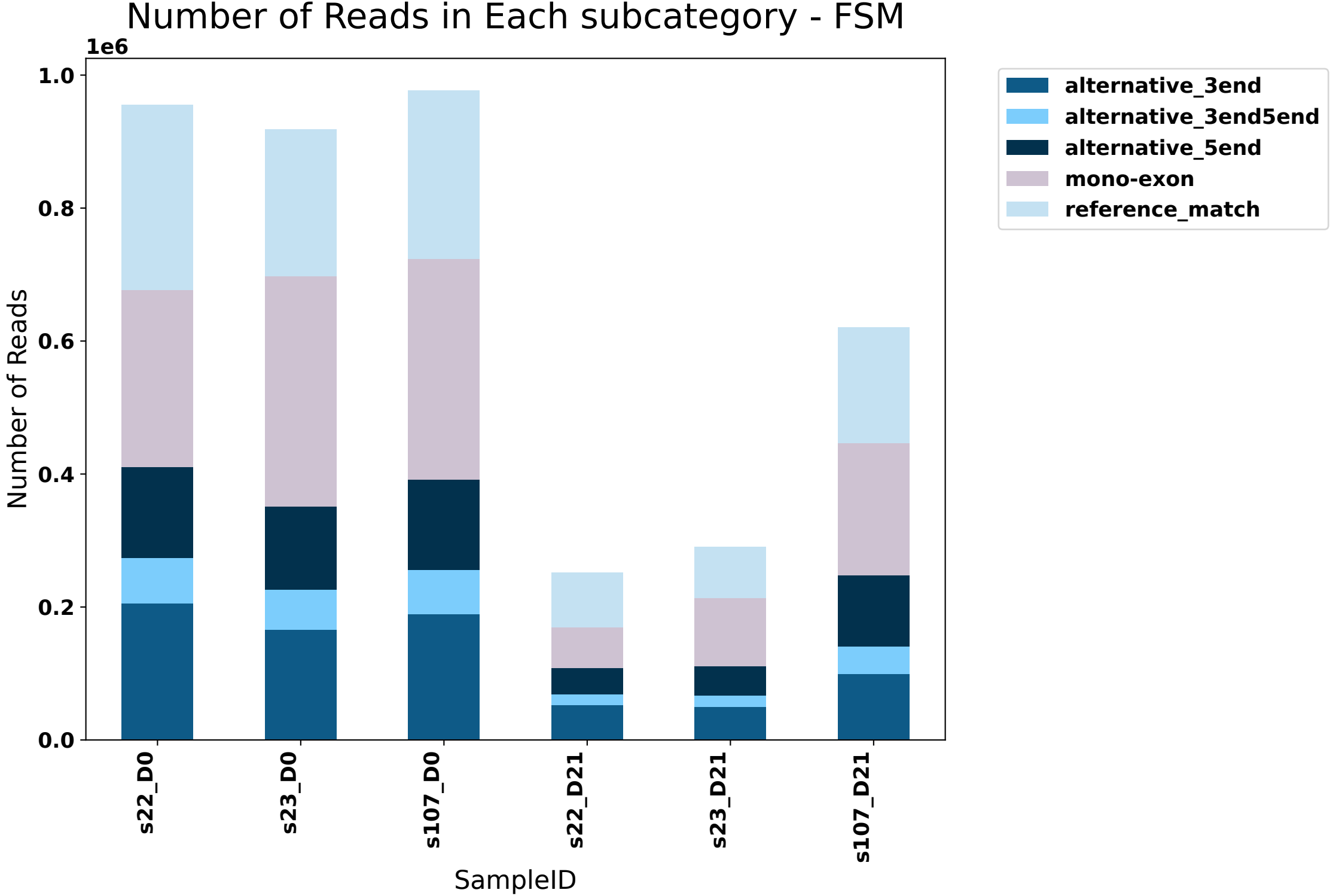
SQANTI-reads report

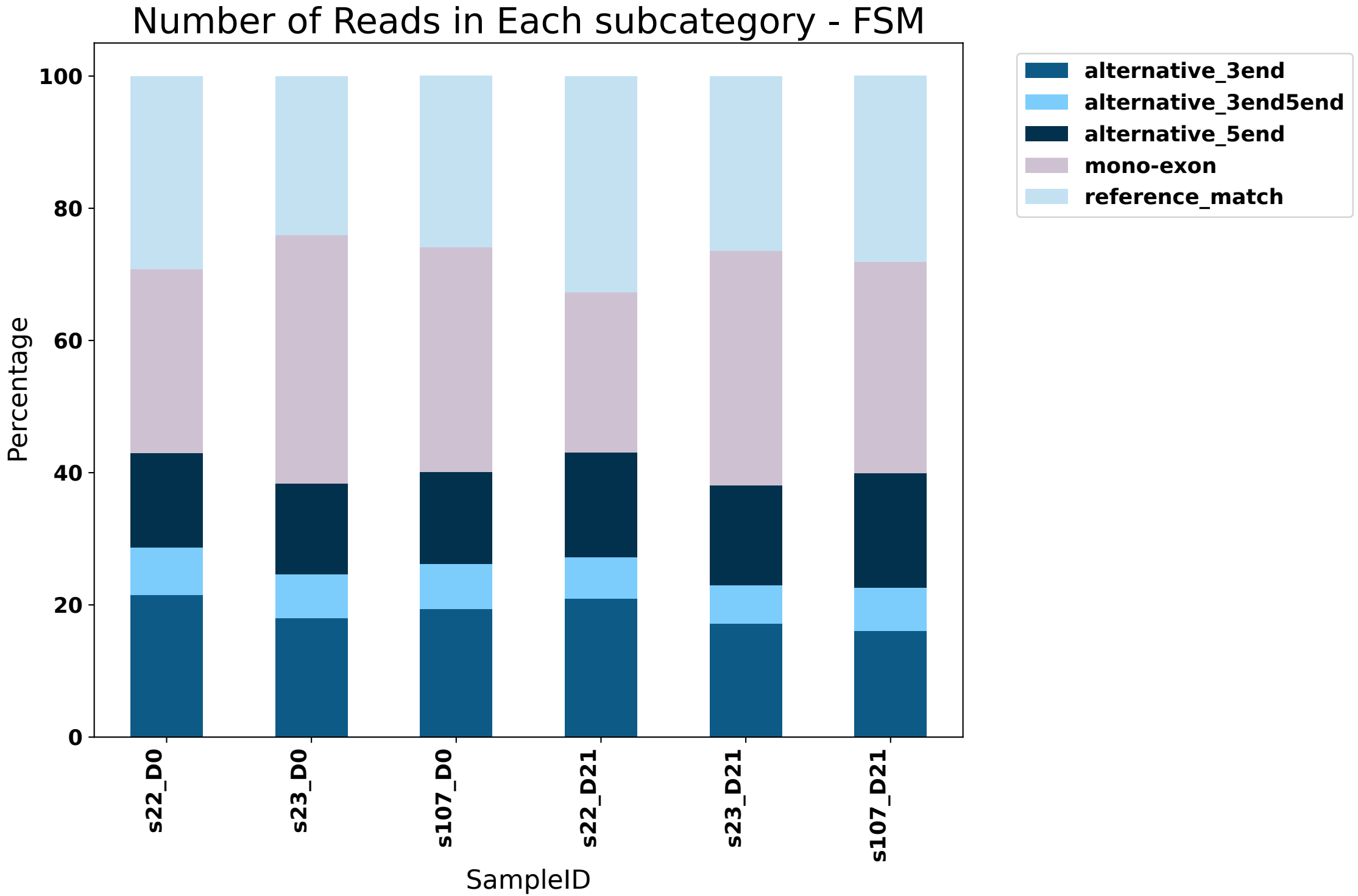

Percent reads in each structural category - All Genes
Percentage
70
60
50
40
30
20
10
0
FSM
ISM
NIC
NNC
AS
FUS
GENIC
GI
INTER
Category
SampleID
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21

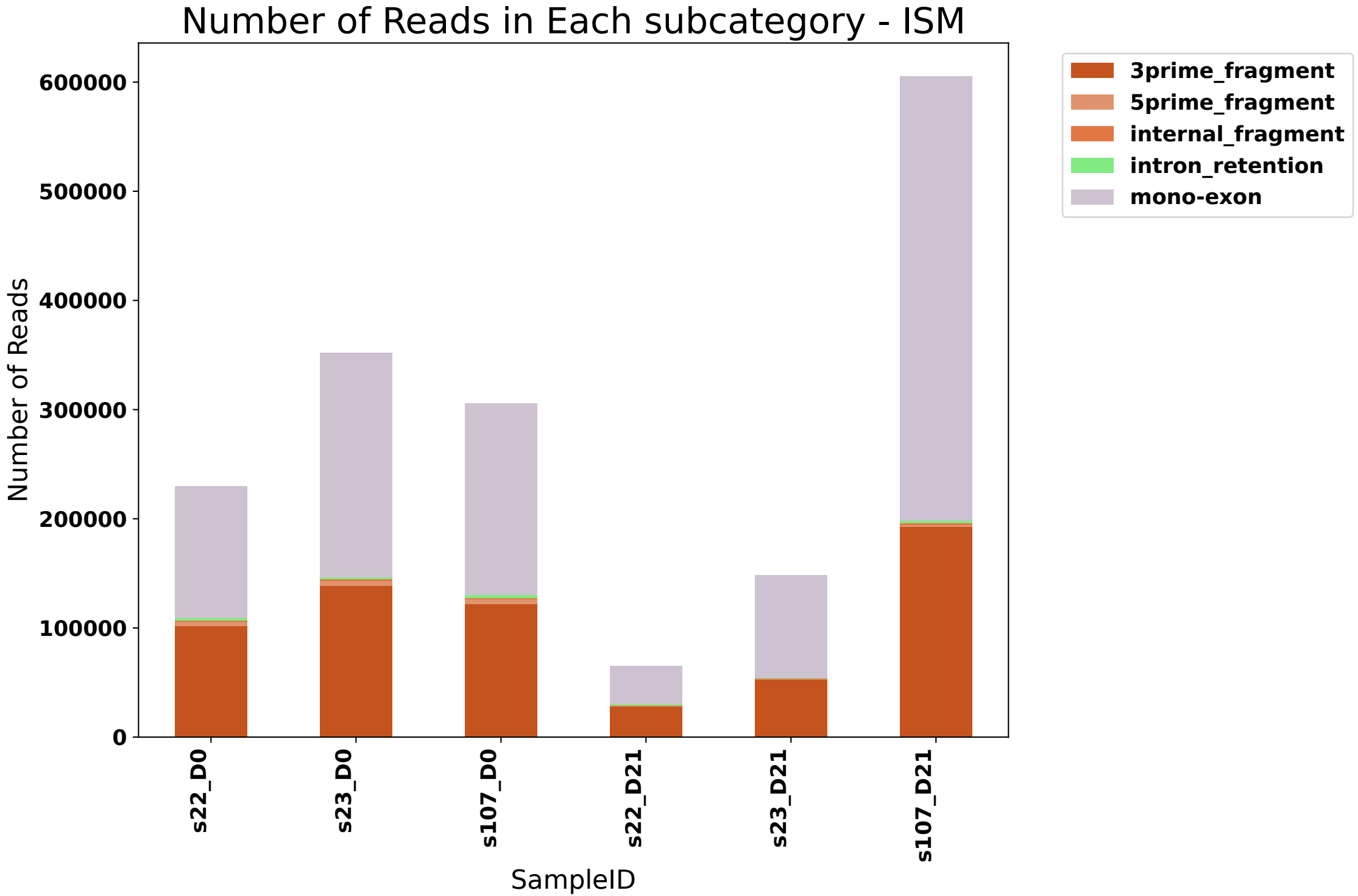

Percent reads in each structural category - Annotated Genes
Percentage
70
60
50
40
30
20
10
0
FSM
ISM
NIC
NNC
AS
FUS
GENIC
GI
INTER
Category
SampleID
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21

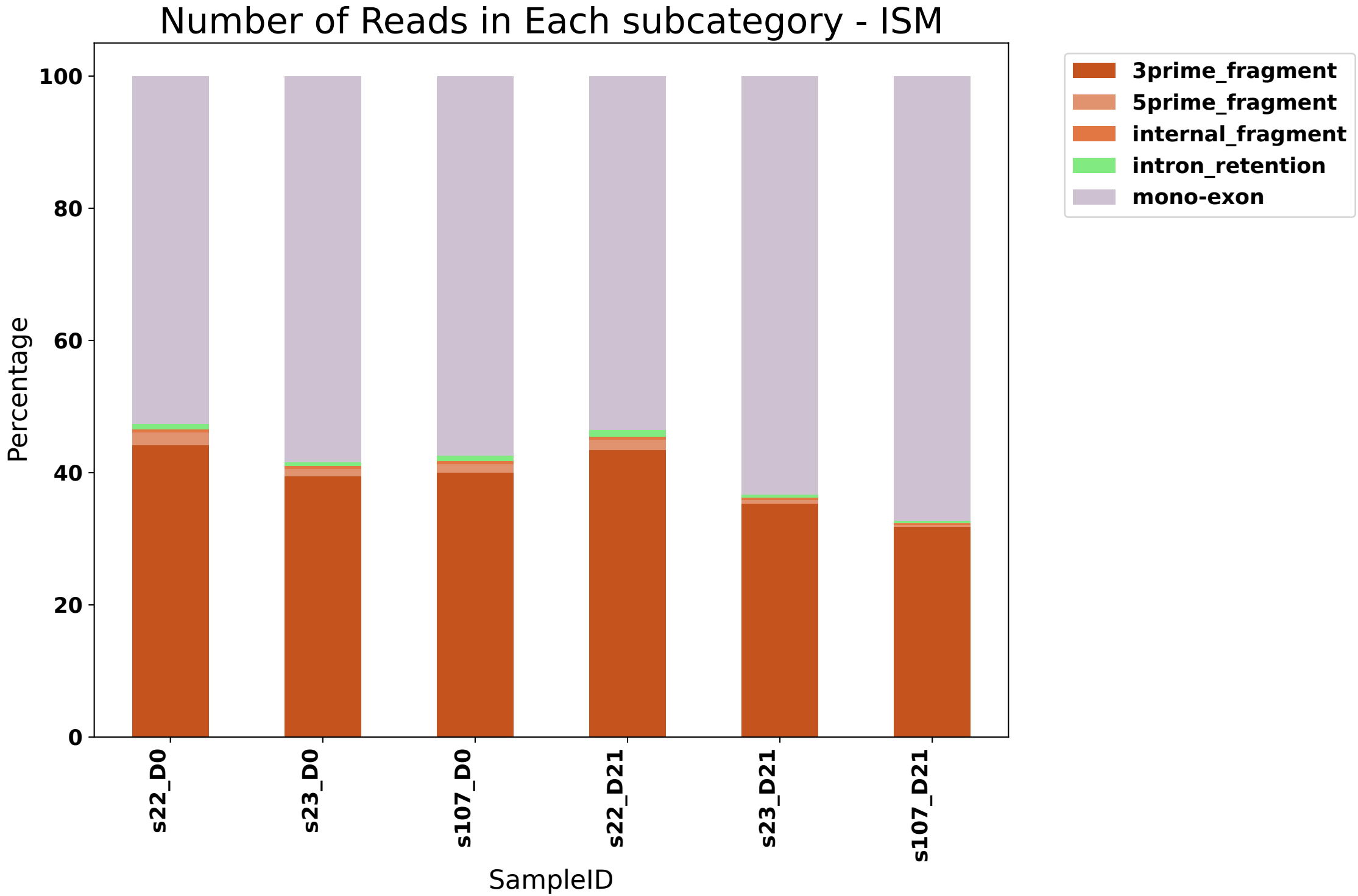

Percent Reads in Each Structural Category - All Genes
Percentages
100
80
60
40
20
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID
INTER
GI
GENIC
FUS
AS
NNC
NIC
ISM
FSM

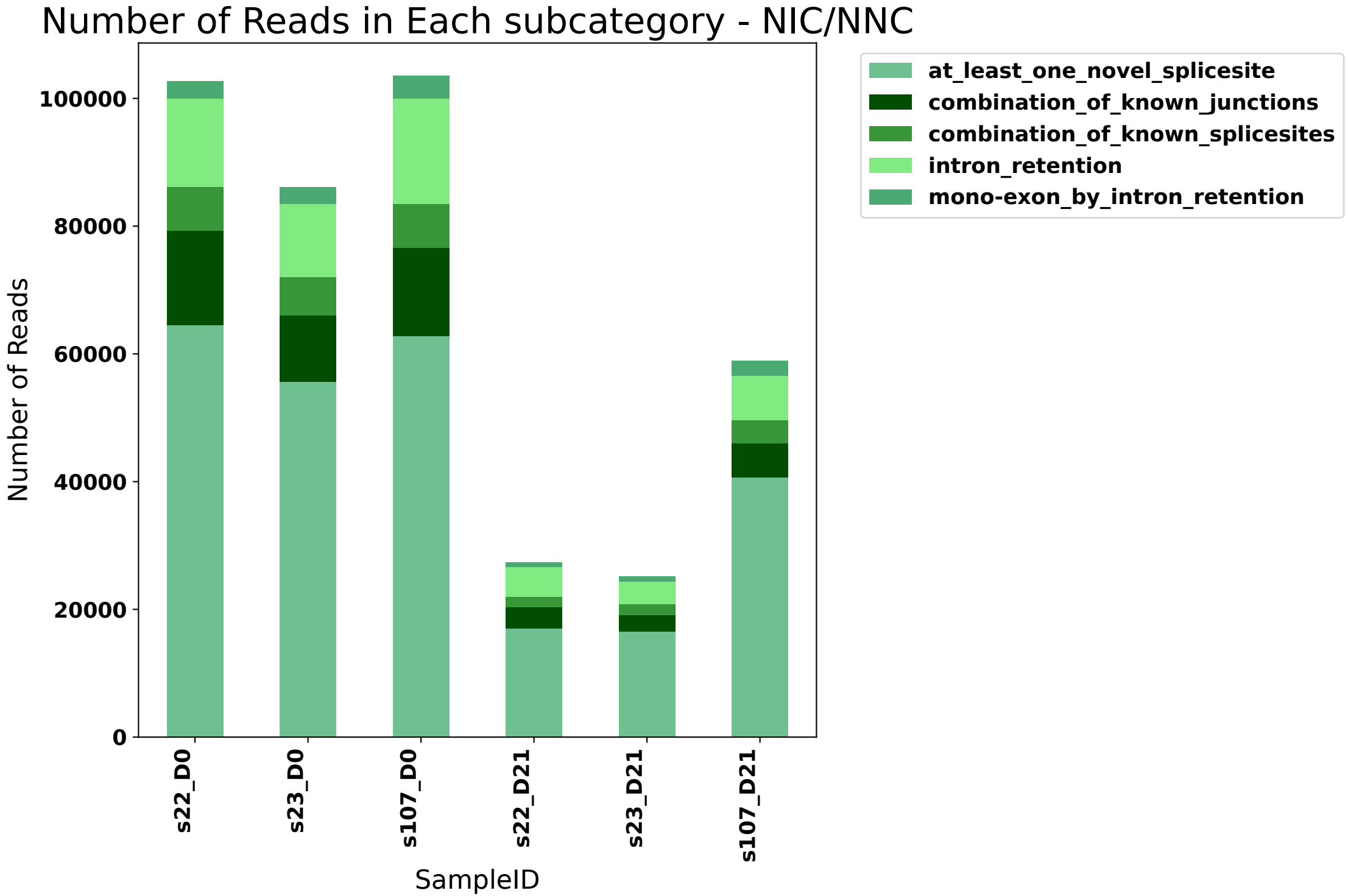

Percent Reads in Each Structural Category - Annotated Genes
Percentages
SampleID
0
20
40
60
80
100
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
GI
GENIC
NNC
NIC
ISM
FSM

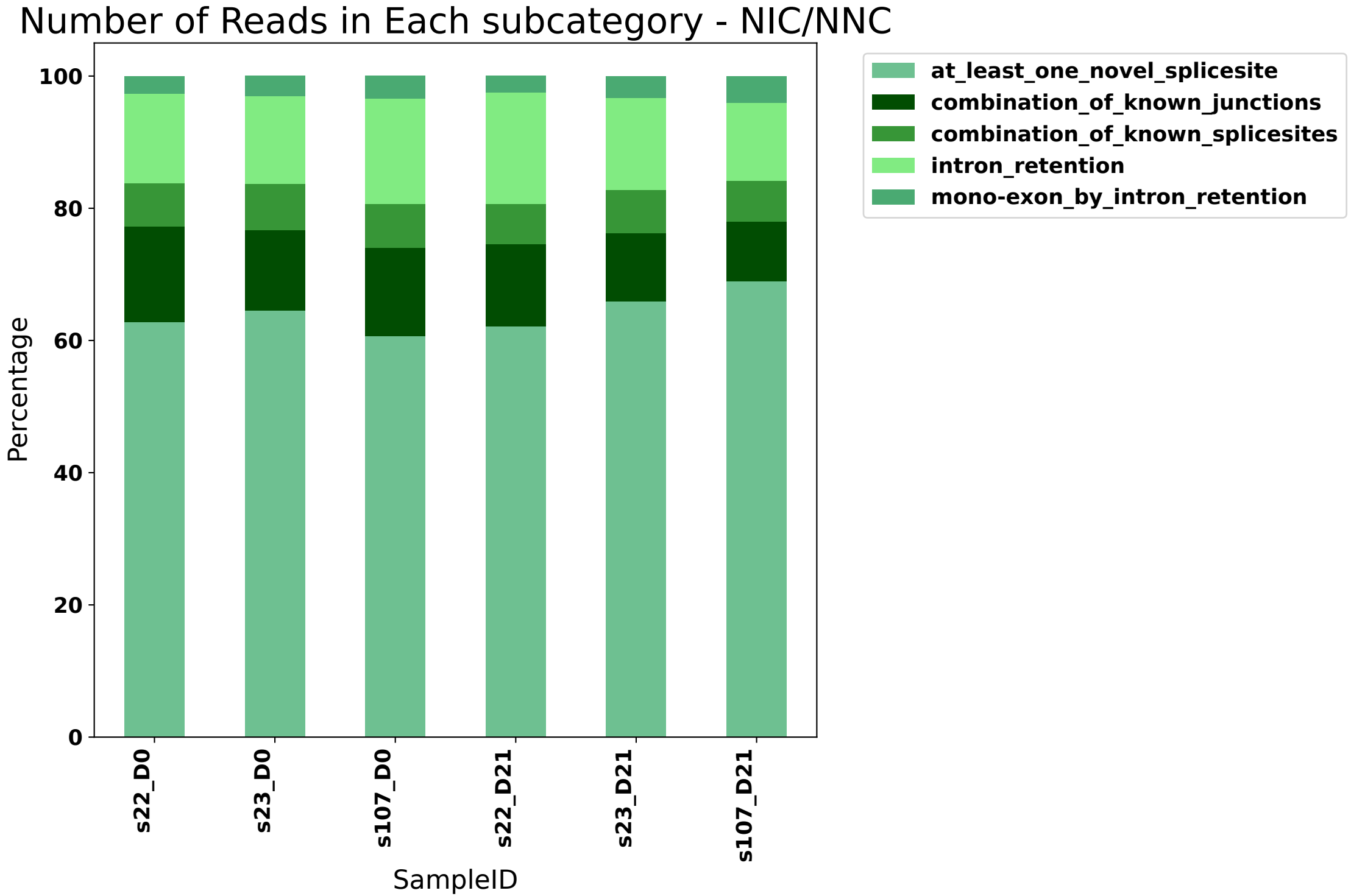

Number of Reads in Each subcategory - FSM
1e6
1.0
0.8
0.6
0.4
0.2
0.0
Number of Reads
s22_D0
s23_D0
s107_D0
s22_D21
s23_D21
s107_D21
SampleID
alternative_3end
alternative_3end5end
alternative_5end
mono-exon
reference_match

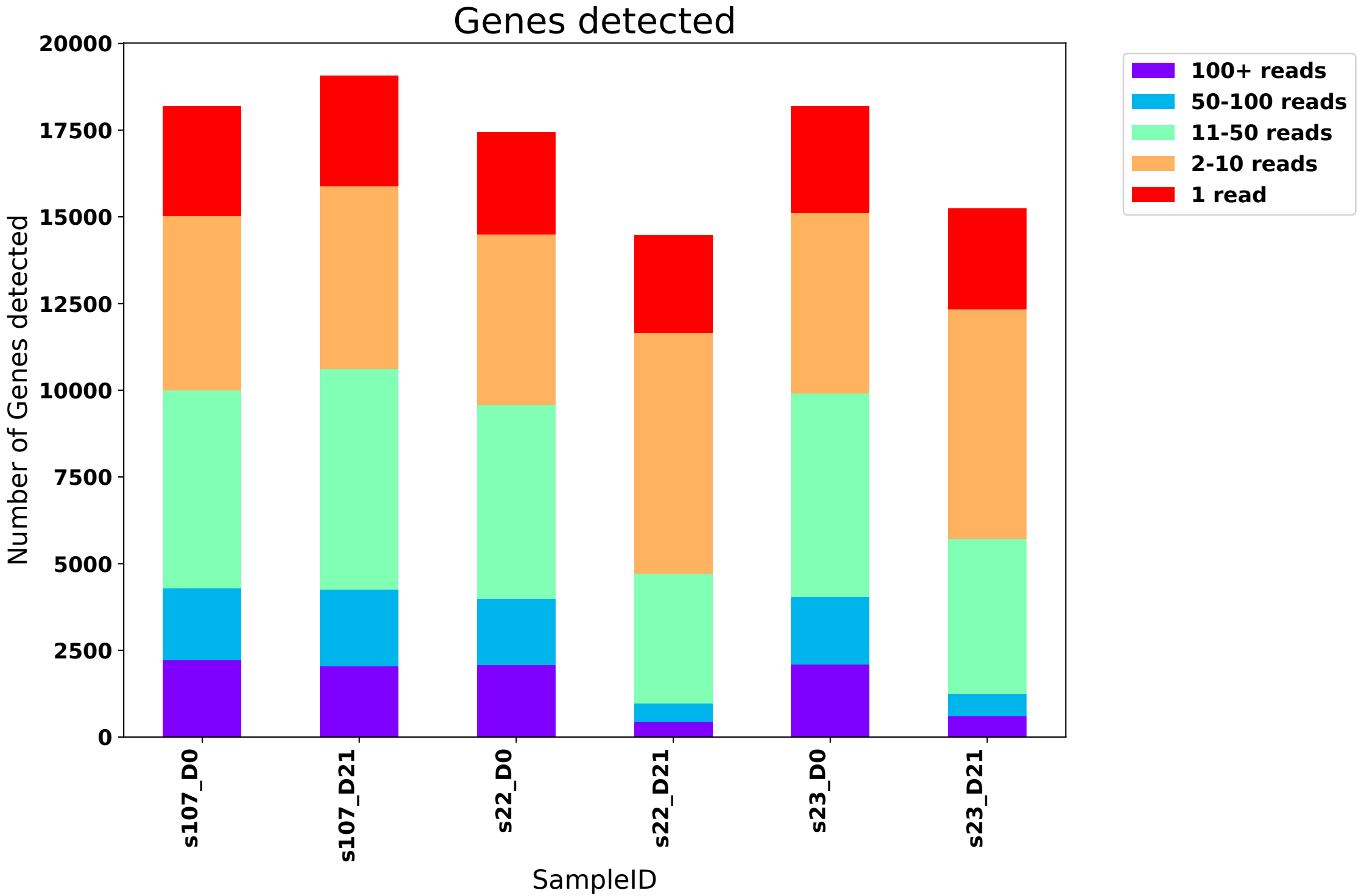

Number of Reads in Each subcategory - FSM
Percentage
SampleID
0
20
40
60
80
100
s22_D0
s23_D0
s107_D0
s22_D21
s23_D21
s107_D21
alternative_3end
alternative_3end5end
alternative_5end
mono-exon
reference_match

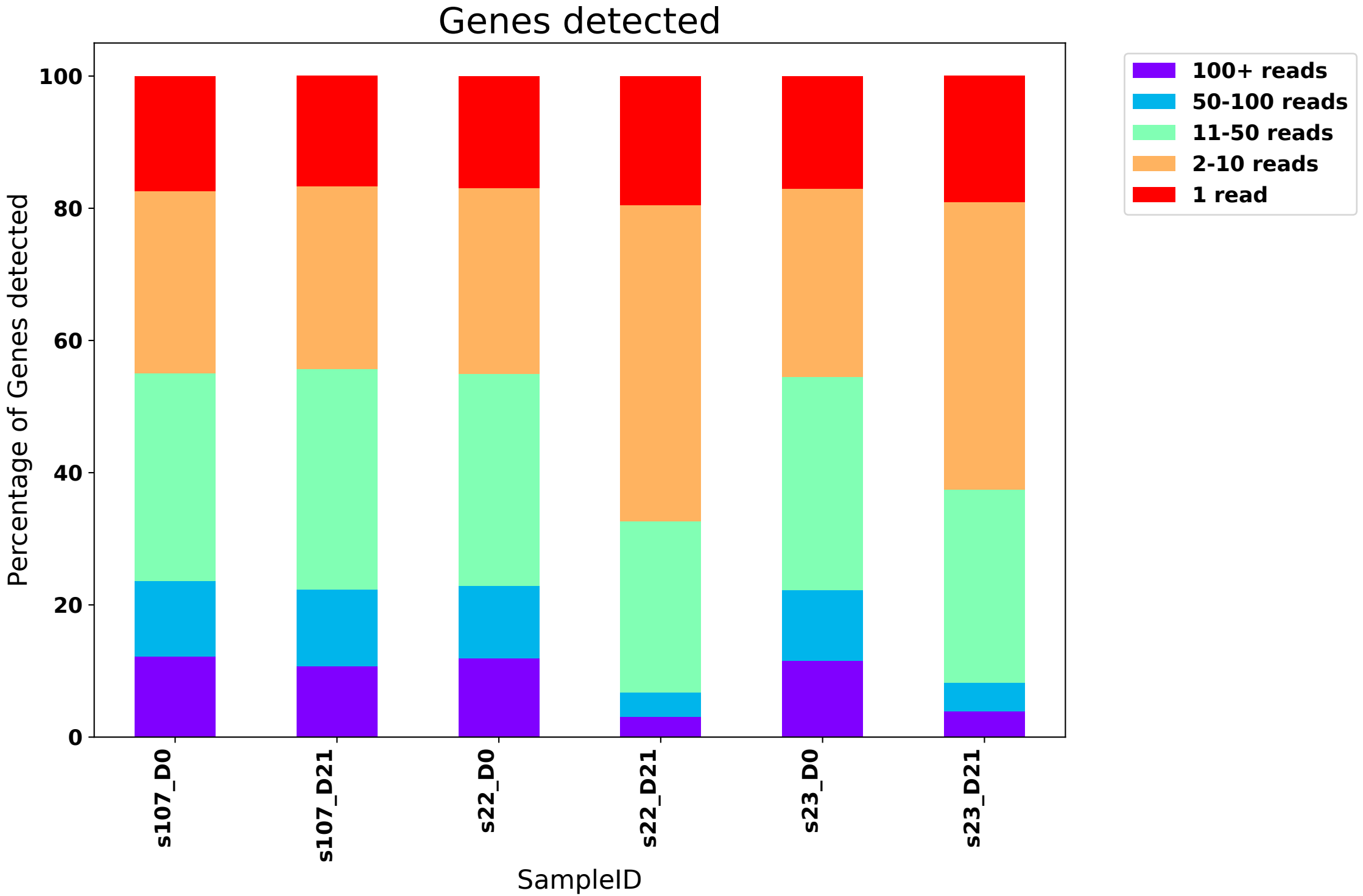

Number of Reads in Each subcategory - ISM
Number of Reads
600000
500000
400000
300000
200000
100000
0
s22_D0
s23_D0
s107_D0
s22_D21
s23_D21
s107_D21
SampleID
3prime_fragment
5prime_fragment
internal_fragment
intron_retention
mono-exon

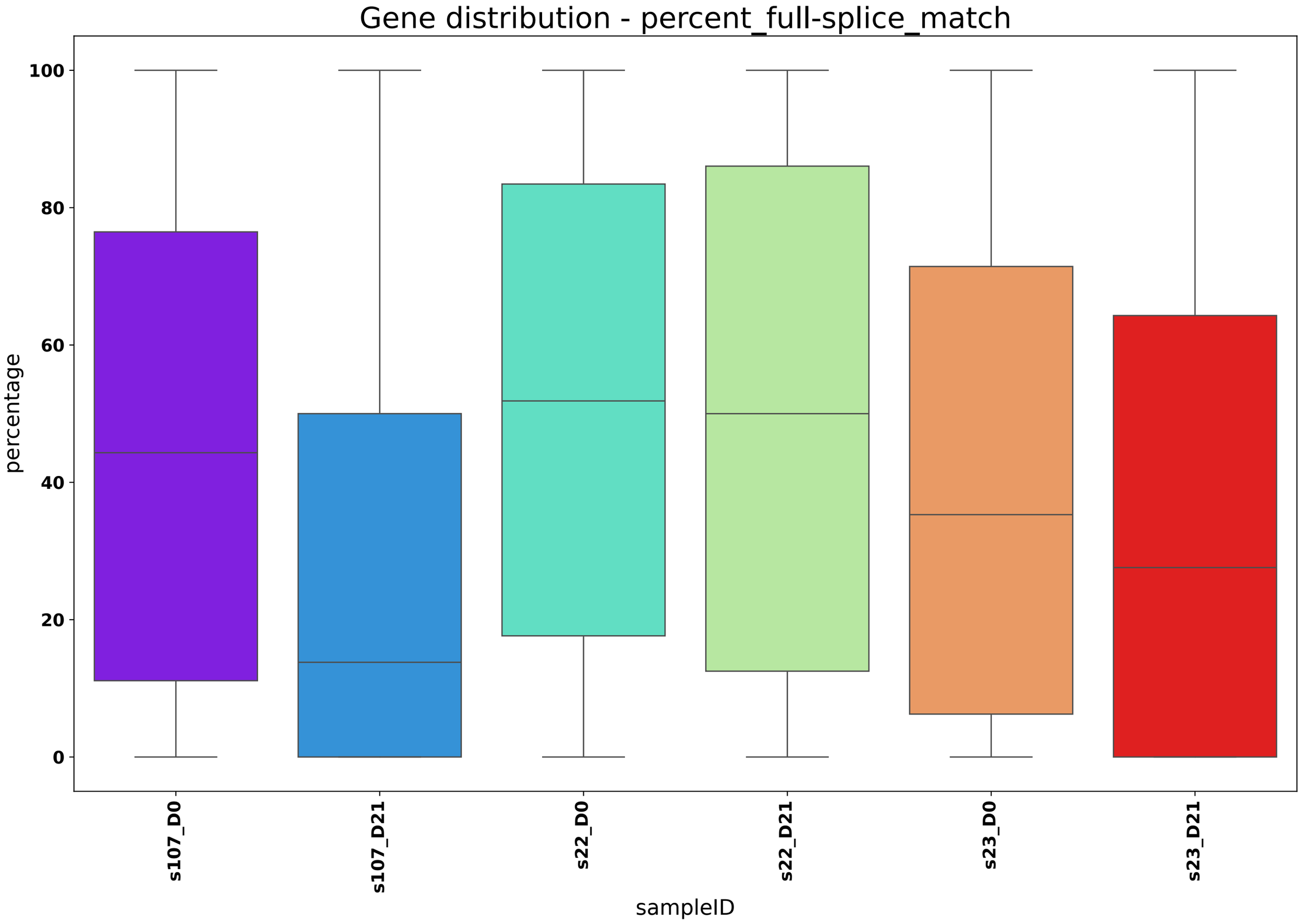

Number of Reads in Each subcategory - ISM
Percentage
SampleID
0
20
40
60
80
100
s22_D0
s23_D0
s107_D0
s22_D21
s23_D21
s107_D21
3prime_fragment
5prime_fragment
internal_fragment
intron_retention
mono-exon

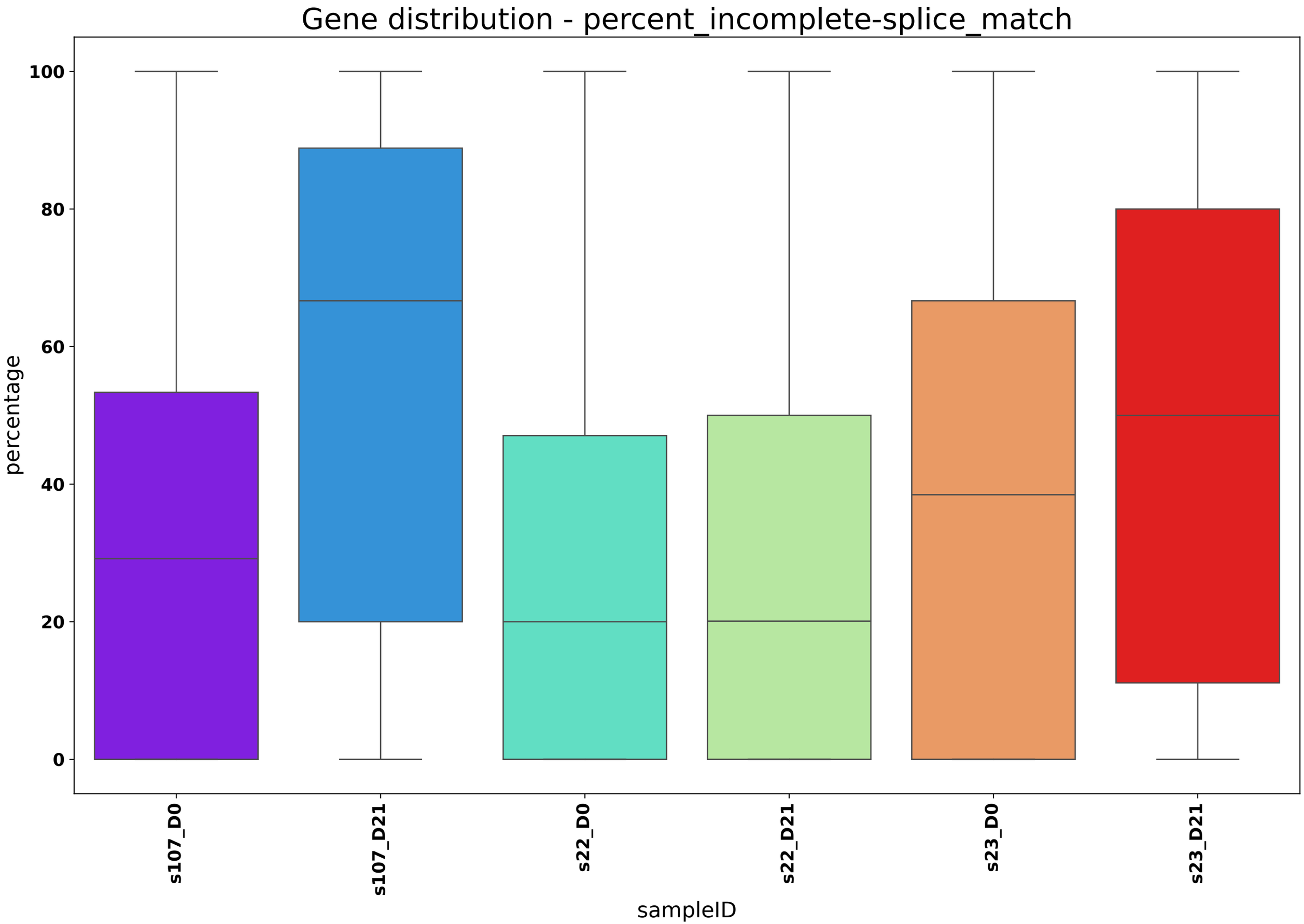

Number of Reads in Each subcategory - NIC/NNC
Number of Reads
100000
80000
60000
40000
20000
0
s22_D0
s23_D0
s107_D0
s22_D21
s23_D21
s107_D21
SampleID
at_least_one_novel_splicesite
combination_of_known_junctions
combination_of_known_splicesites
intron_retention
mono-exon_by_intron_retention

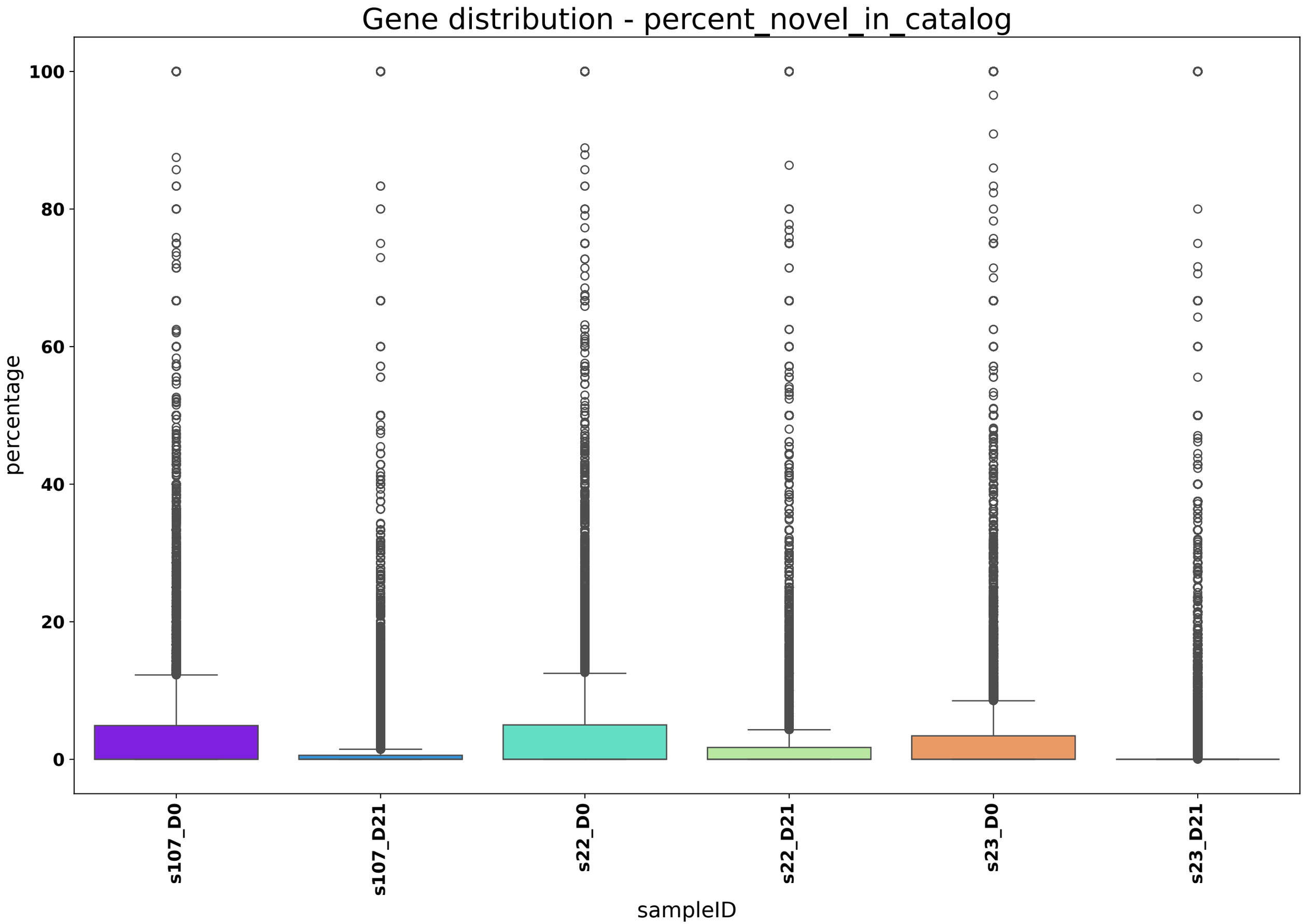

Number of Reads in Each subcategory - NIC/NNC
Percentage
100
80
60
40
20
0
s22_D0
s23_D0
s107_D0
s22_D21
s23_D21
s107_D21
SampleID
at_least_one_novel_splicesite
combination_of_known_junctions
combination_of_known_splicesites
intron_retention
mono-exon_by_intron_retention

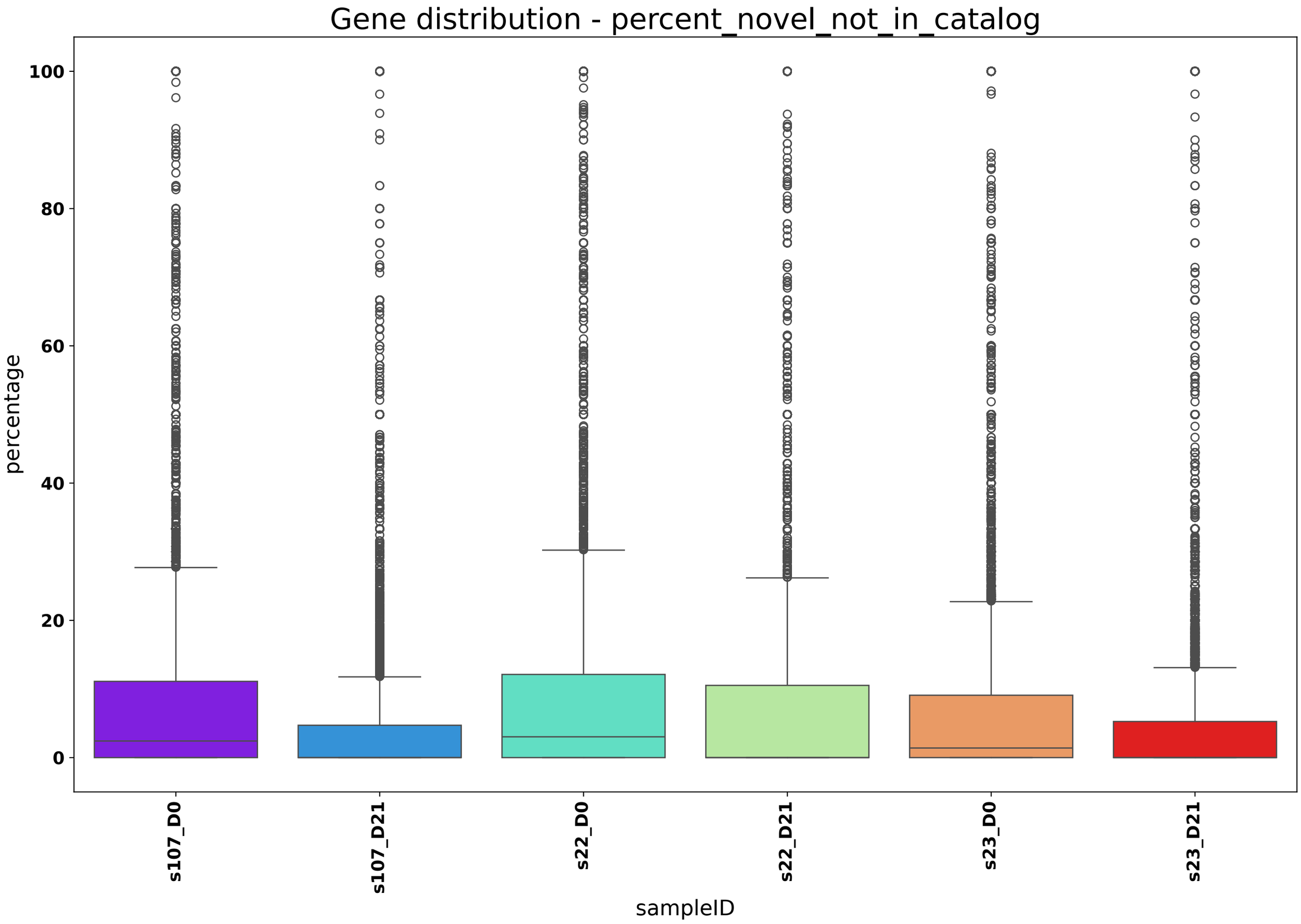

Genes detected
Number of Genes detected
20000
17500
15000
12500
10000
7500
5000
2500
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID
100+ reads
50-100 reads
11-50 reads
2-10 reads
1 read

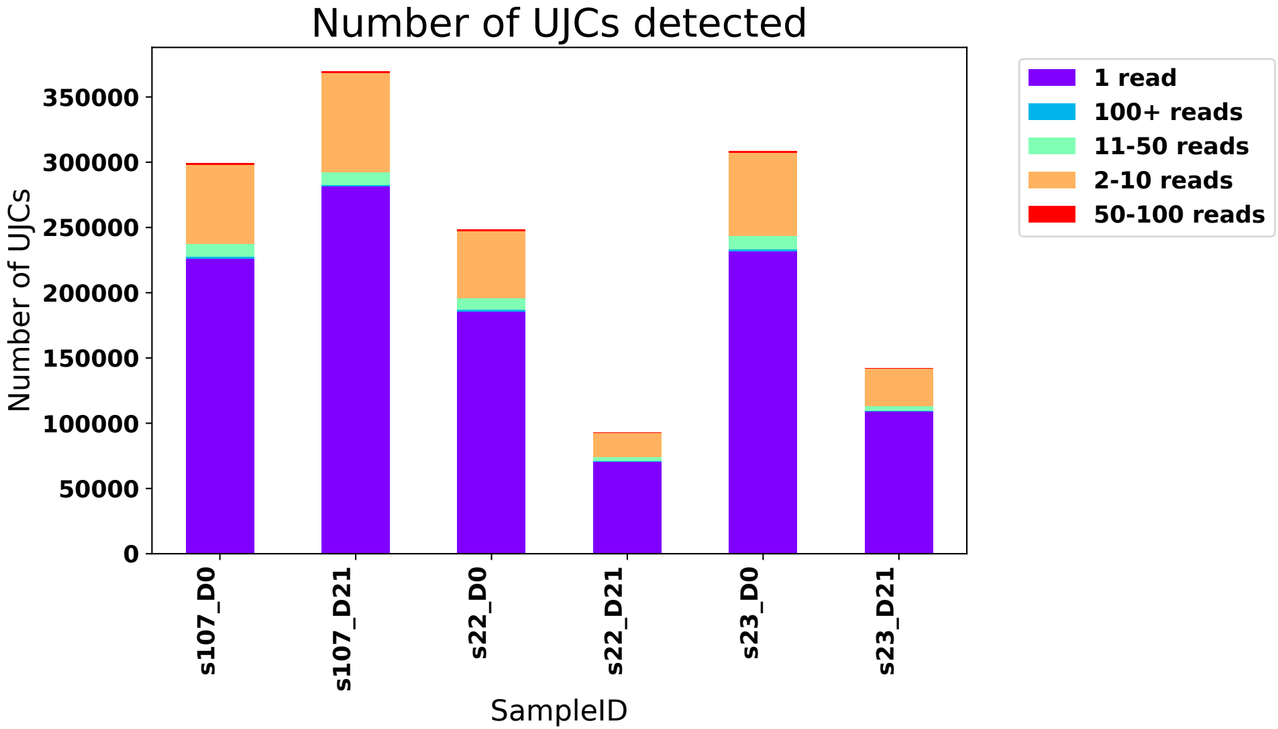

Genes detected
Percentage of Genes detected
100
80
60
40
20
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID
100+ reads
50-100 reads
11-50 reads
2-10 reads
1 read

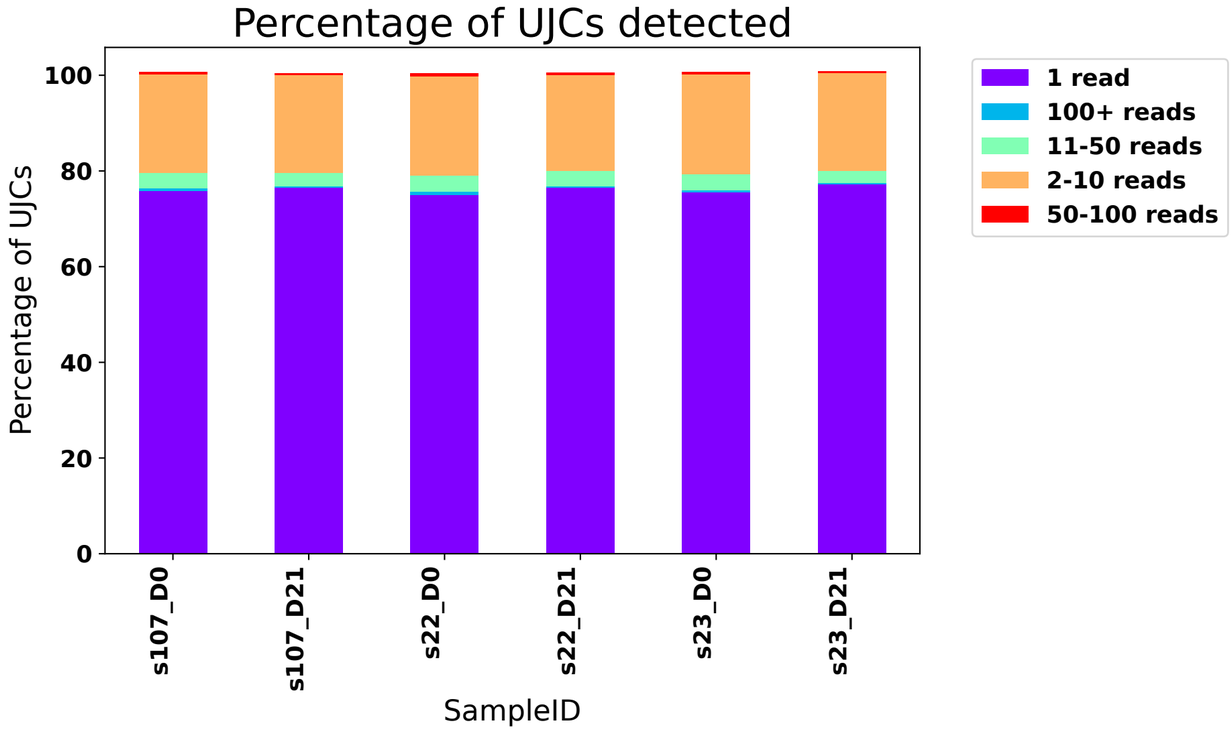

Gene distribution - percent_full-splice_match
percentage
100
80
60
40
20
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
sampleID

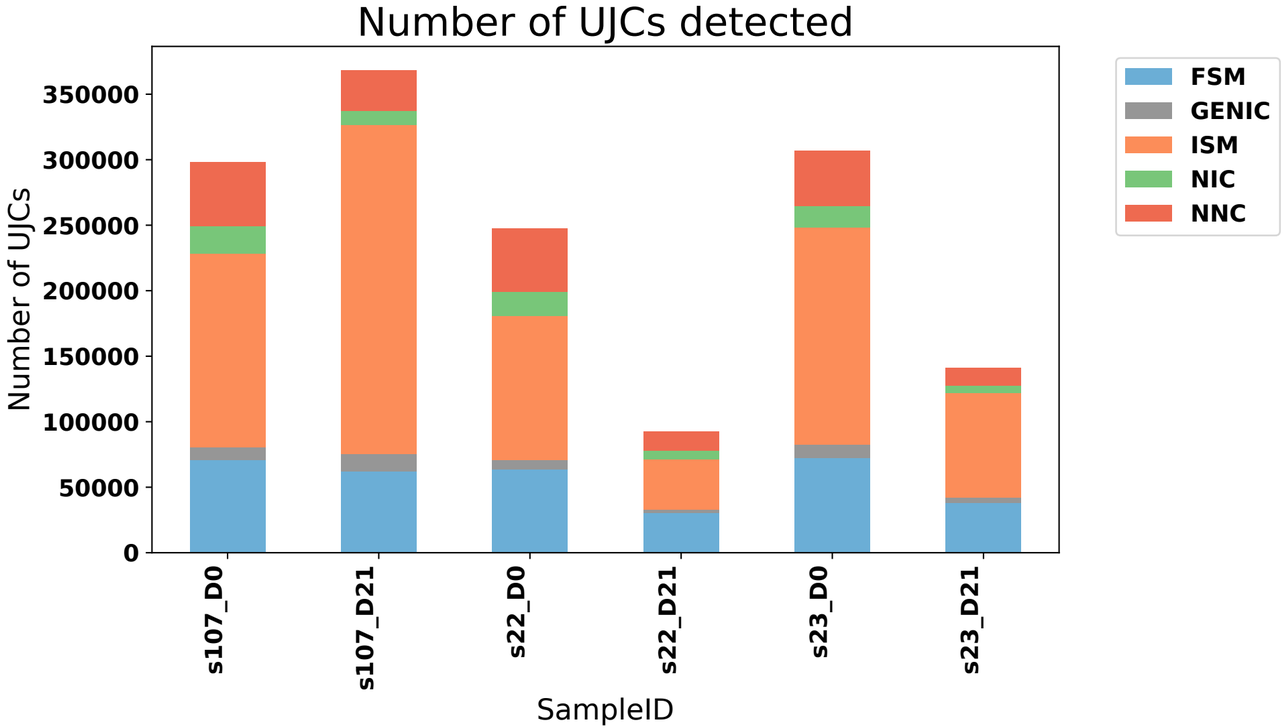

Gene distribution - percent_incomplete-splice_match
percentage
100
80
60
40
20
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
sampleID

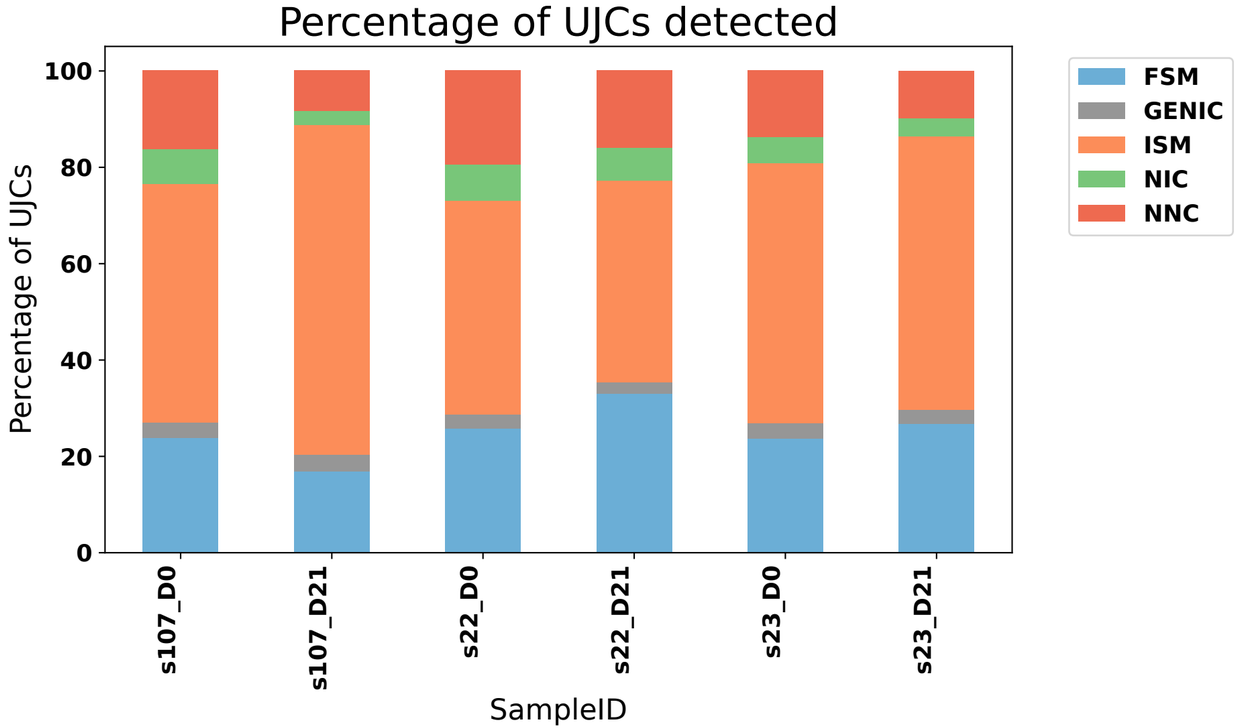

Gene distribution - percent_novel_in_catalog
percentage
100
80
60
40
20
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
sampleID

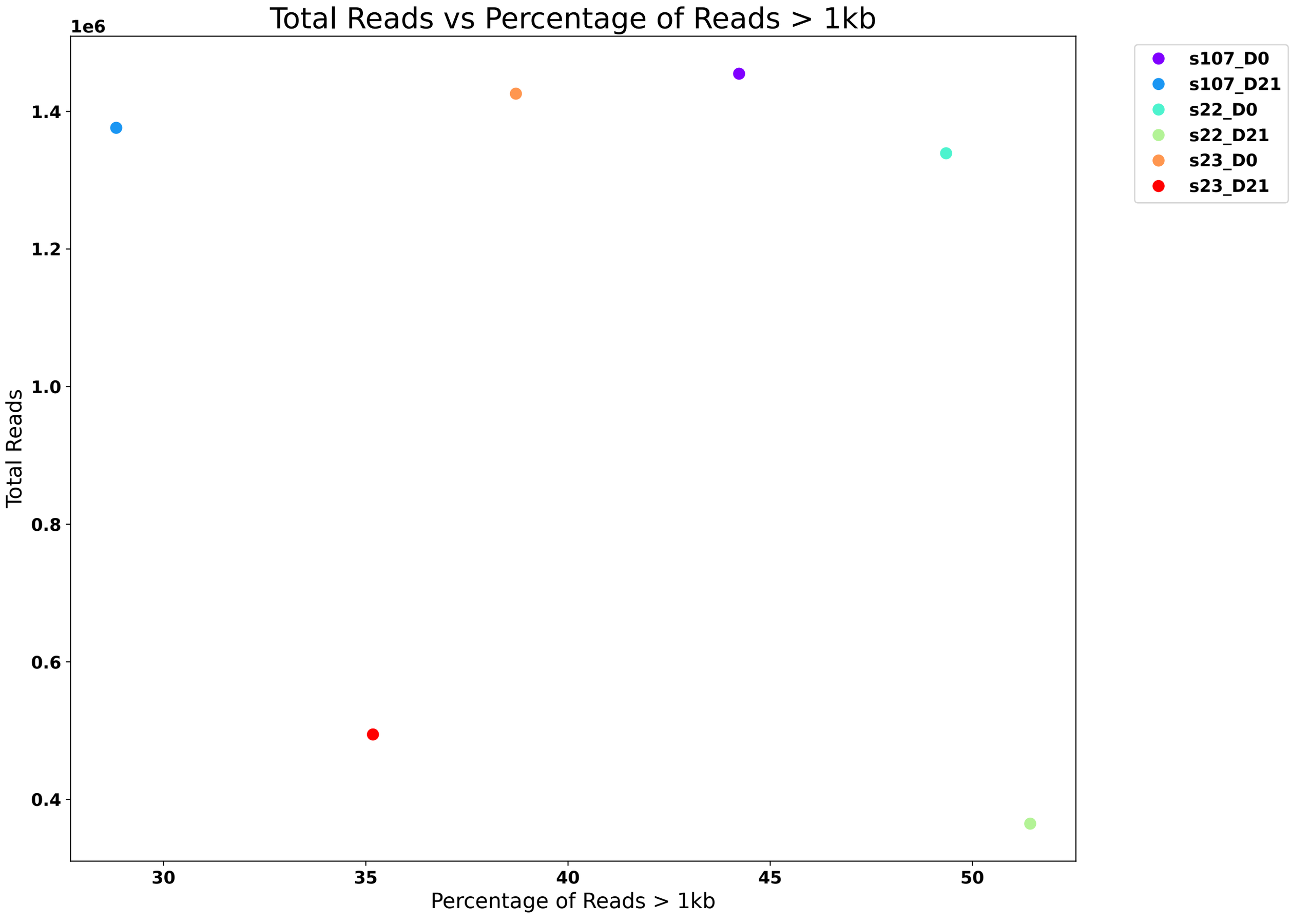

Gene distribution - percent_novel_not_in_catalog
percentage
100
80
60
40
20
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
sampleID

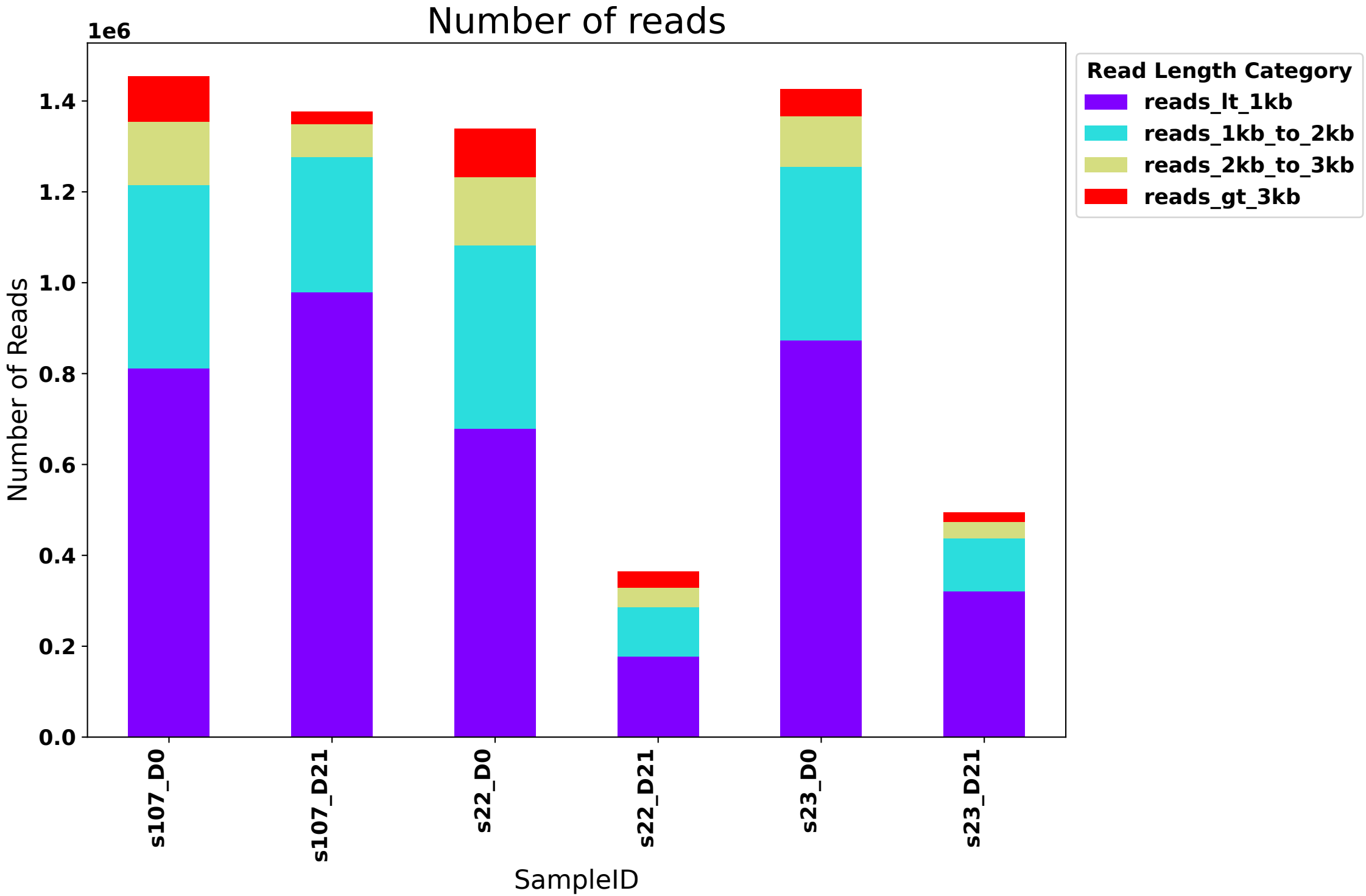

Number of UJCs detected
Number of UJCs
350000
300000
250000
200000
150000
100000
50000
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID
1 read
100+ reads
11-50 reads
2-10 reads
50-100 reads

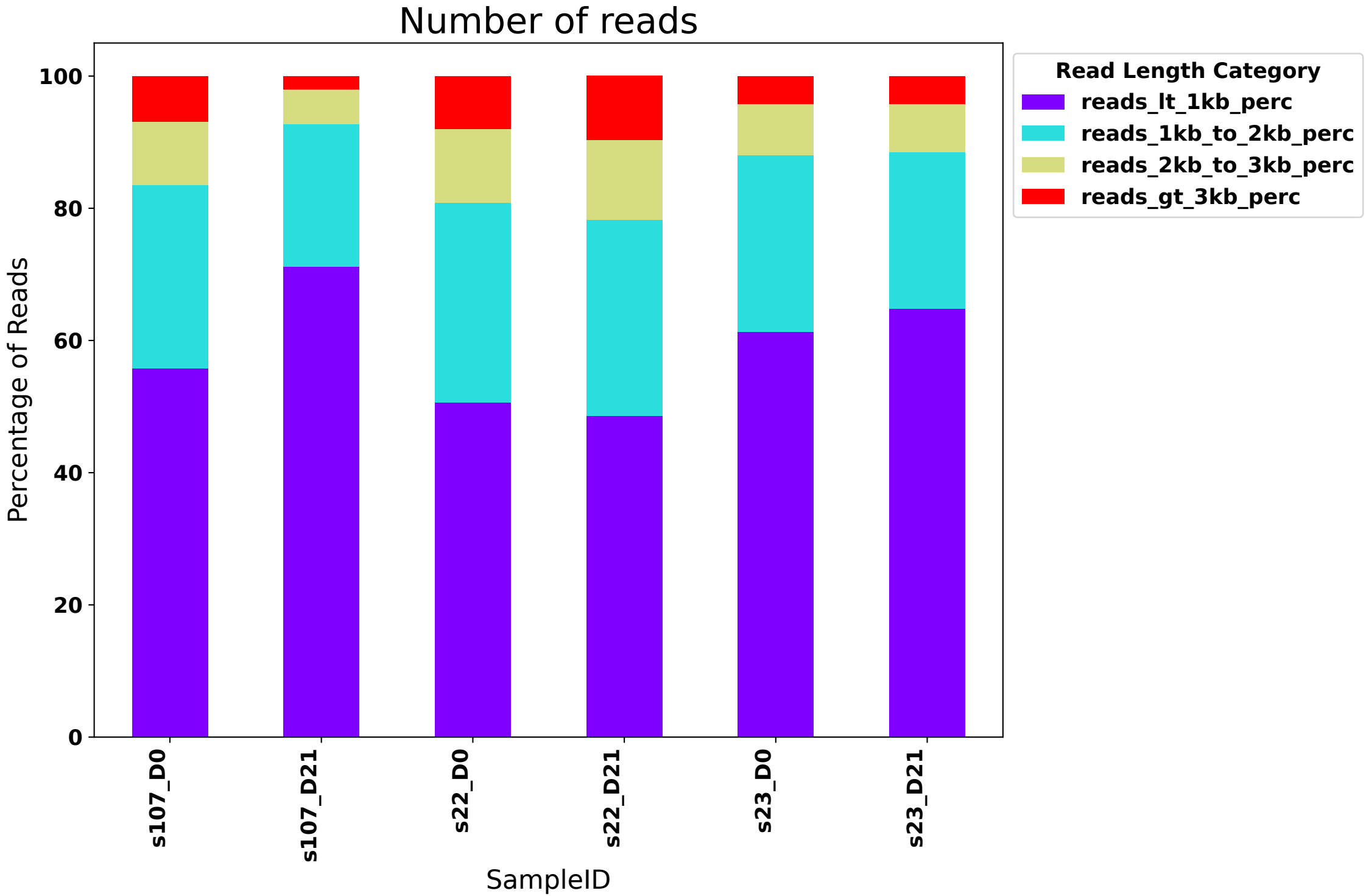

Percentage of UJCs detected
Percentage of UJCs
100
80
60
40
20
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID
1 read
100+ reads
11-50 reads
2-10 reads
50-100 reads

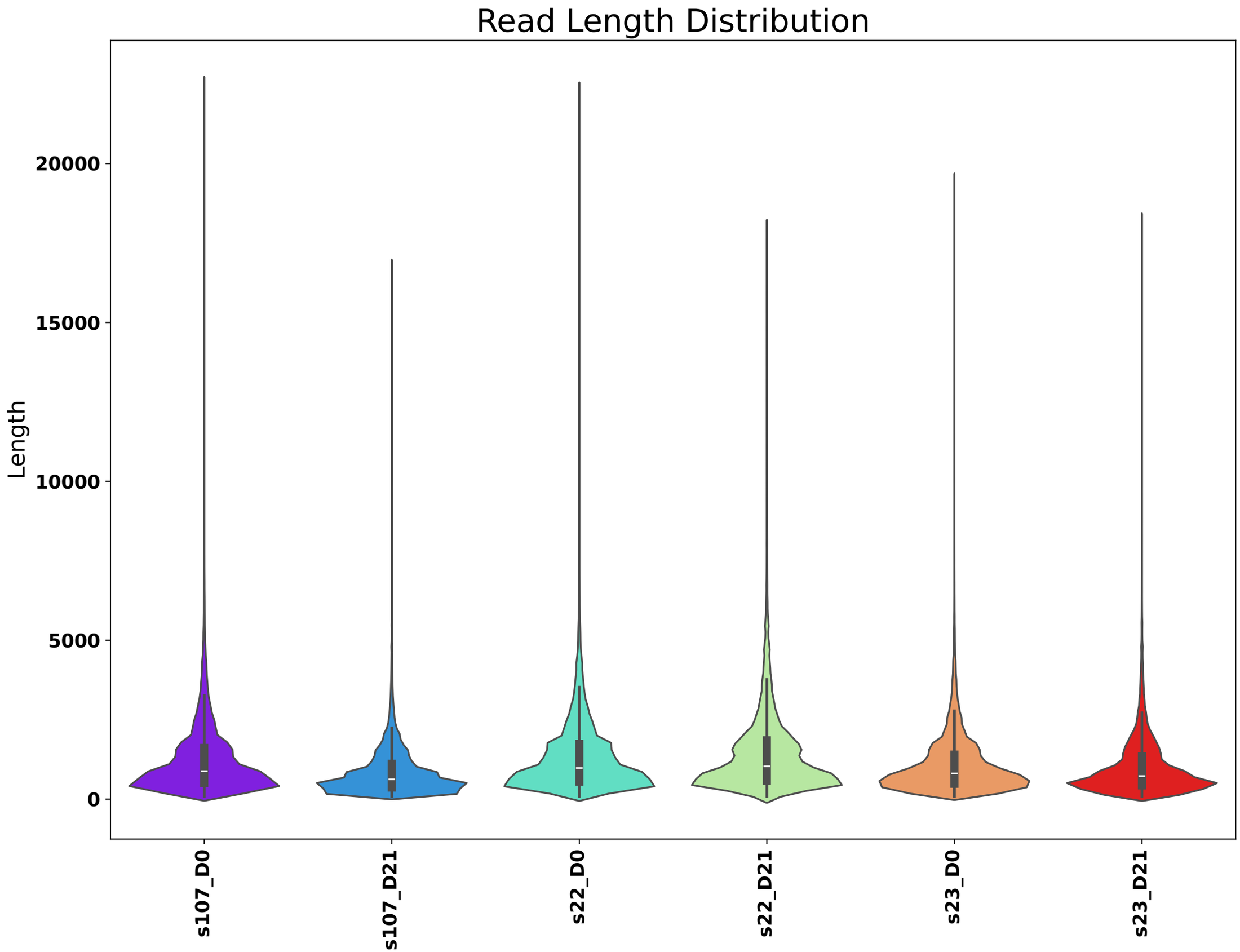

Number of UJCs detected
Number of UJCs
350000
300000
250000
200000
150000
100000
50000
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID
FSM
GENIC
ISM
NIC
NNC

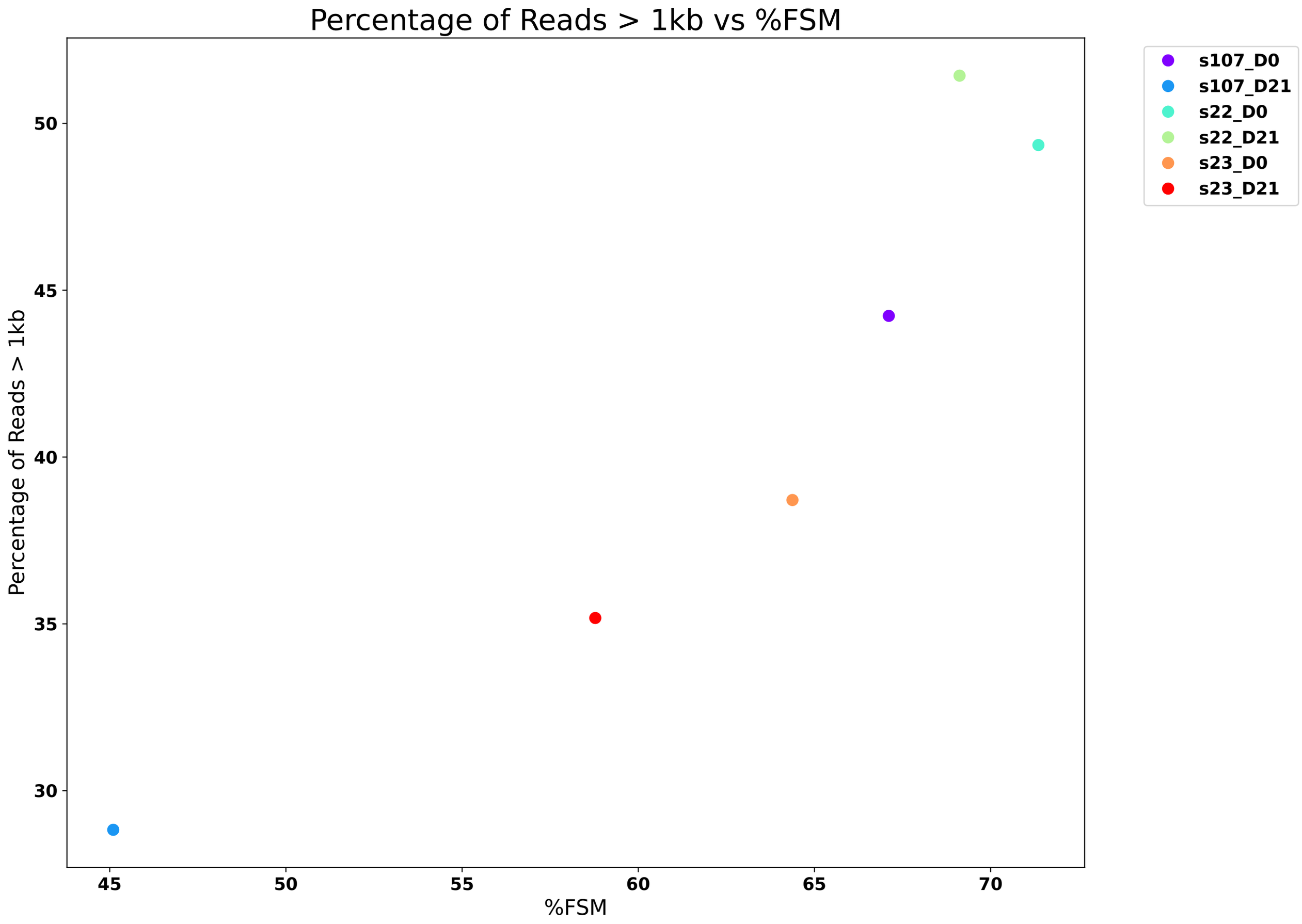

Percentage of UJCs detected
Percentage of UJCs
100
80
60
40
20
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID
FSM
GENIC
ISM
NIC
NNC

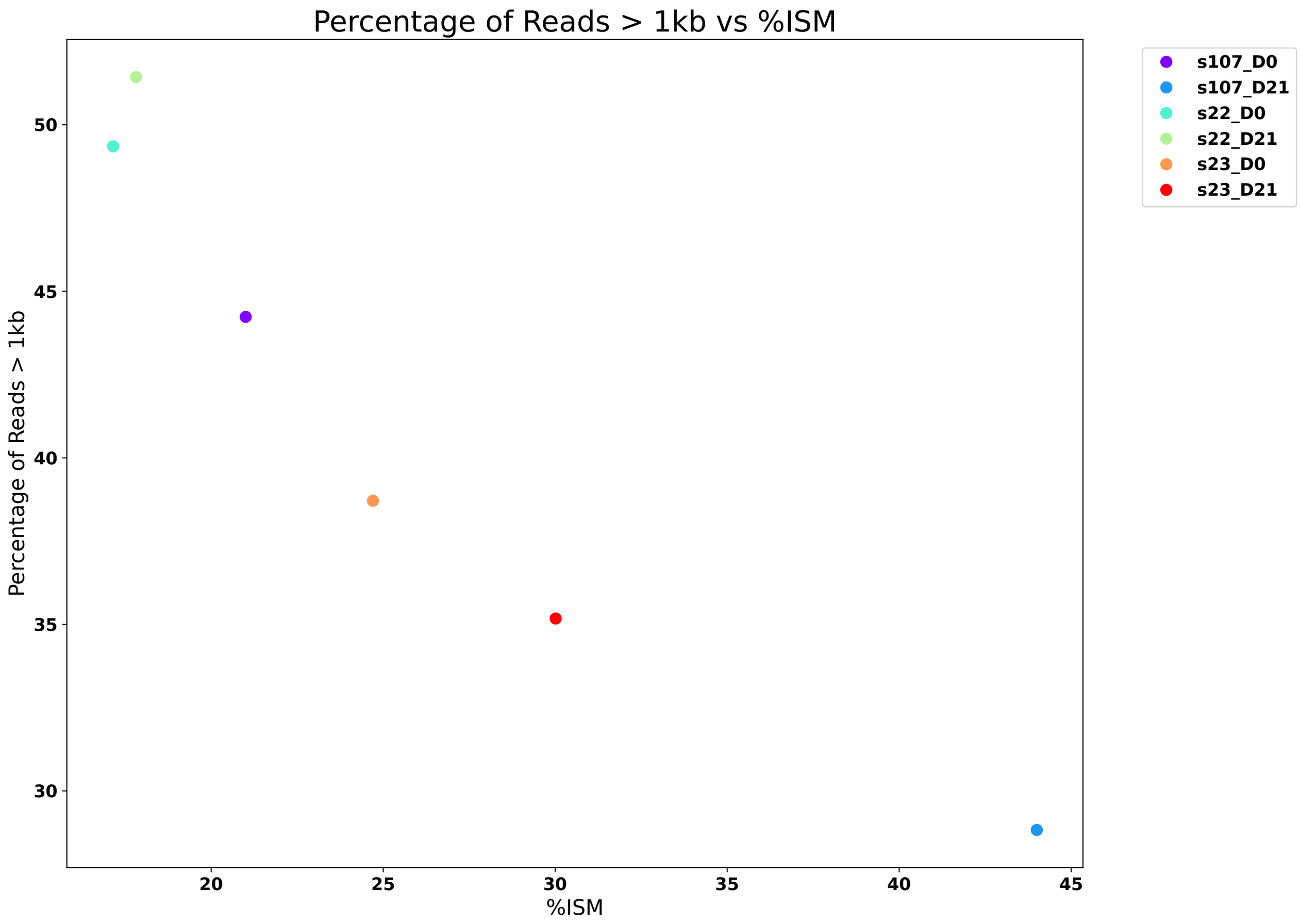

Total Reads vs Percentage of Reads > 1kb
1e6
1.4
1.2
1.0
0.8
0.6
0.4
Total Reads
30
35
40
45
50
Percentage of Reads > 1kb
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21

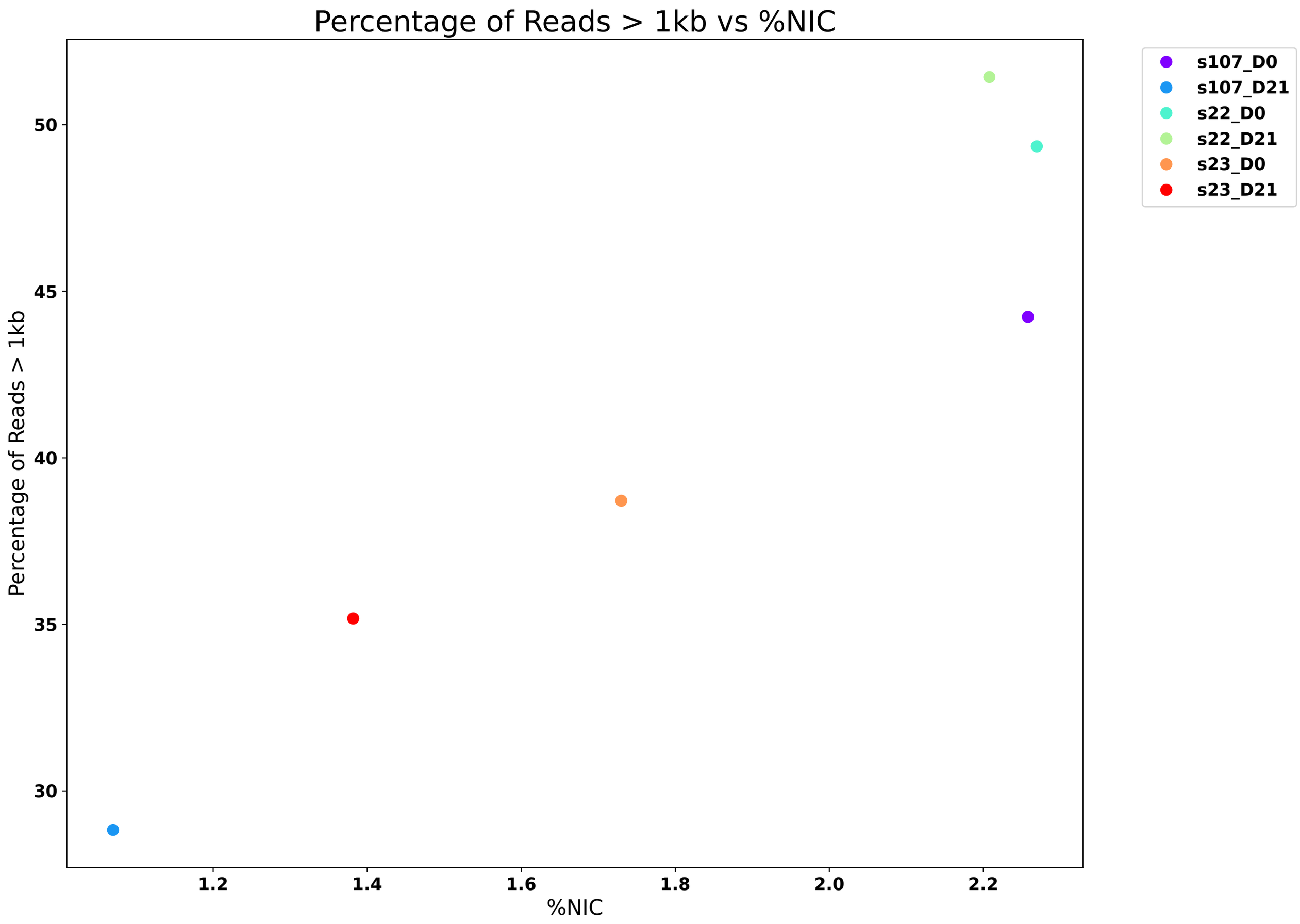

Number of reads
1e6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Number of Reads
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID
Read Length Category
reads_lt_1kb
reads_1kb_to_2kb
reads_2kb_to_3kb
reads_gt_3kb

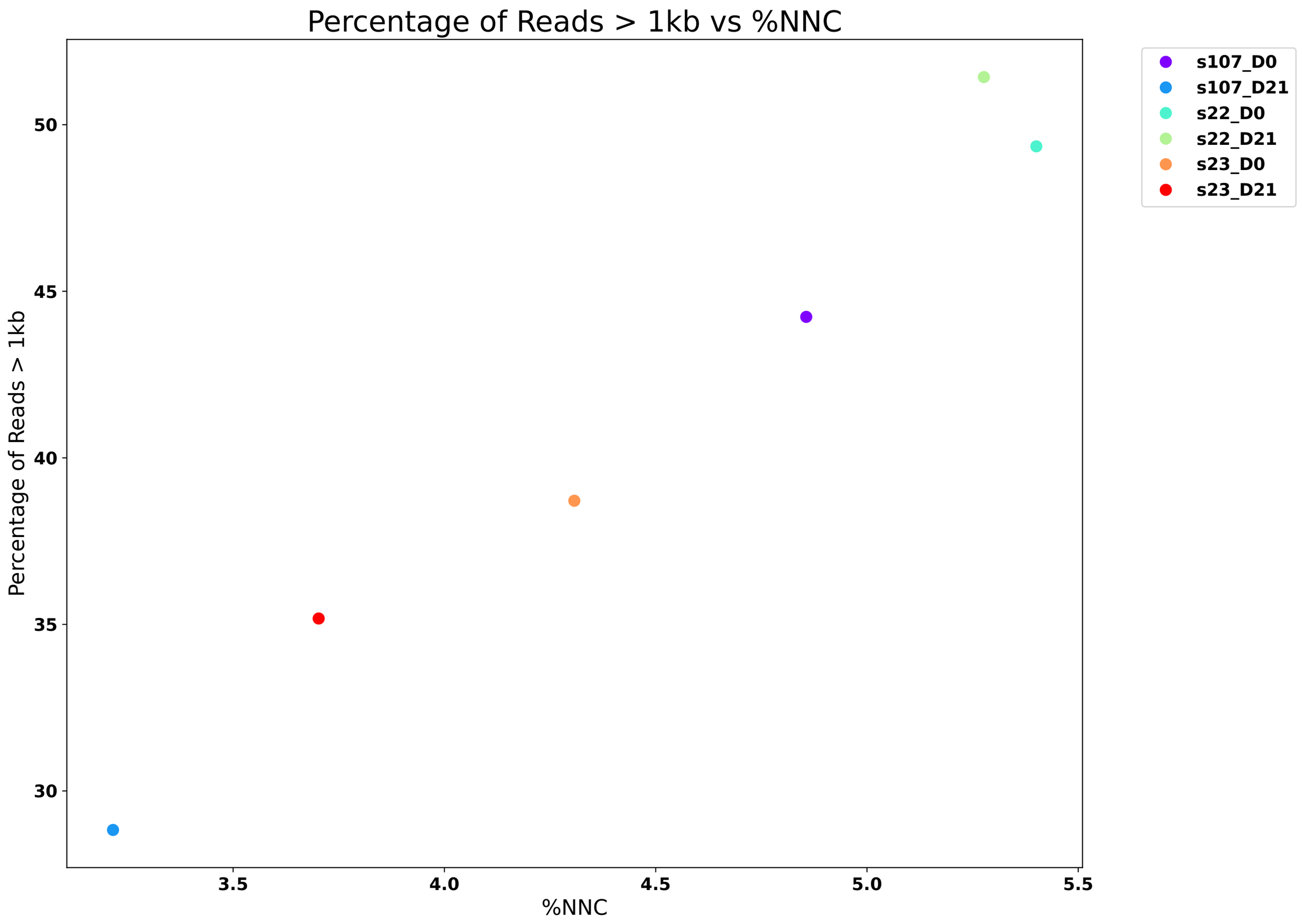

Number of reads
Percentage of Reads
100
80
60
40
20
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID
Read Length Category
reads_lt_1kb_perc
reads_1kb_to_2kb_perc
reads_2kb_to_3kb_perc
reads_gt_3kb_perc

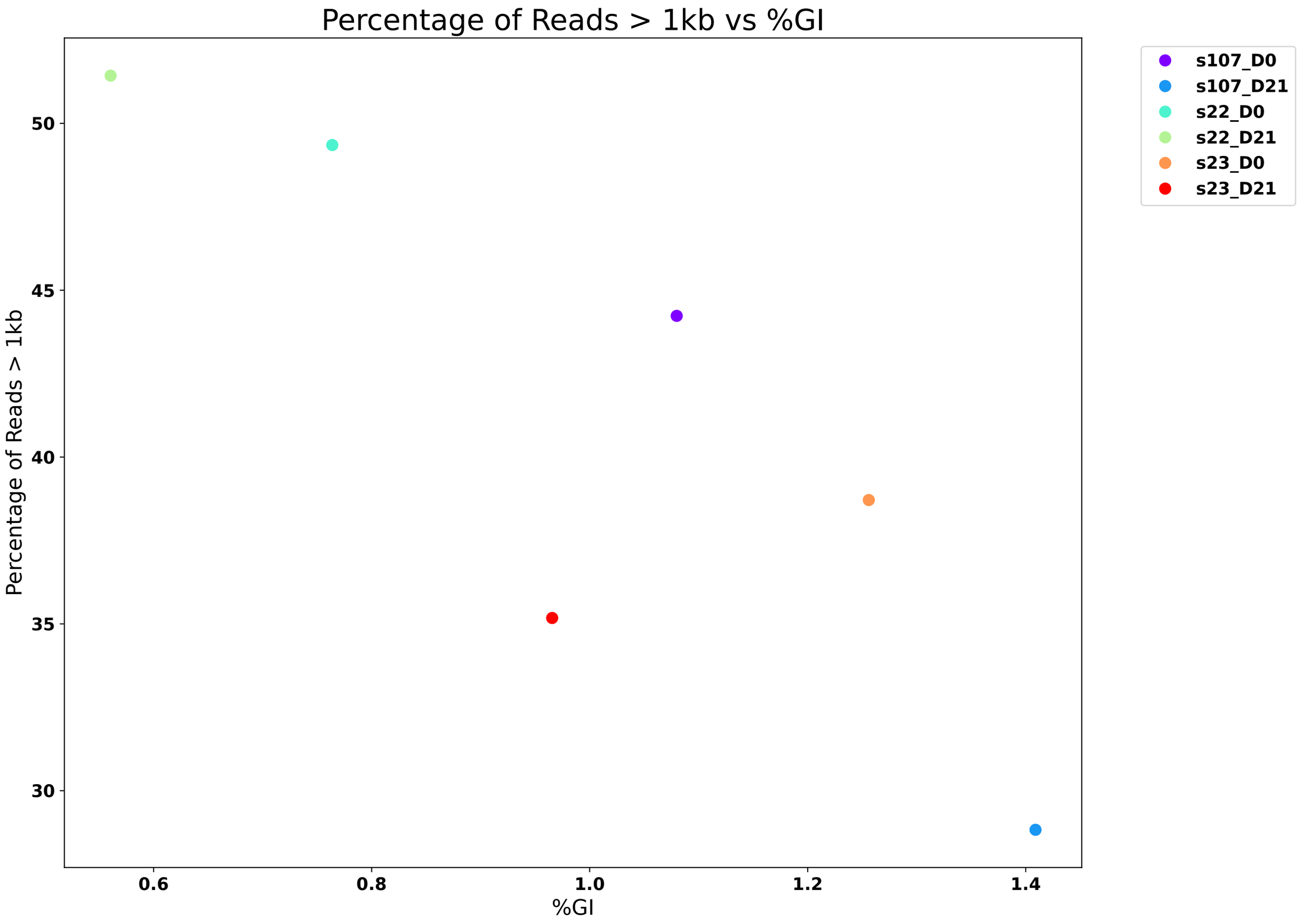

Read Length Distribution
Length
20000
15000
10000
5000
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21

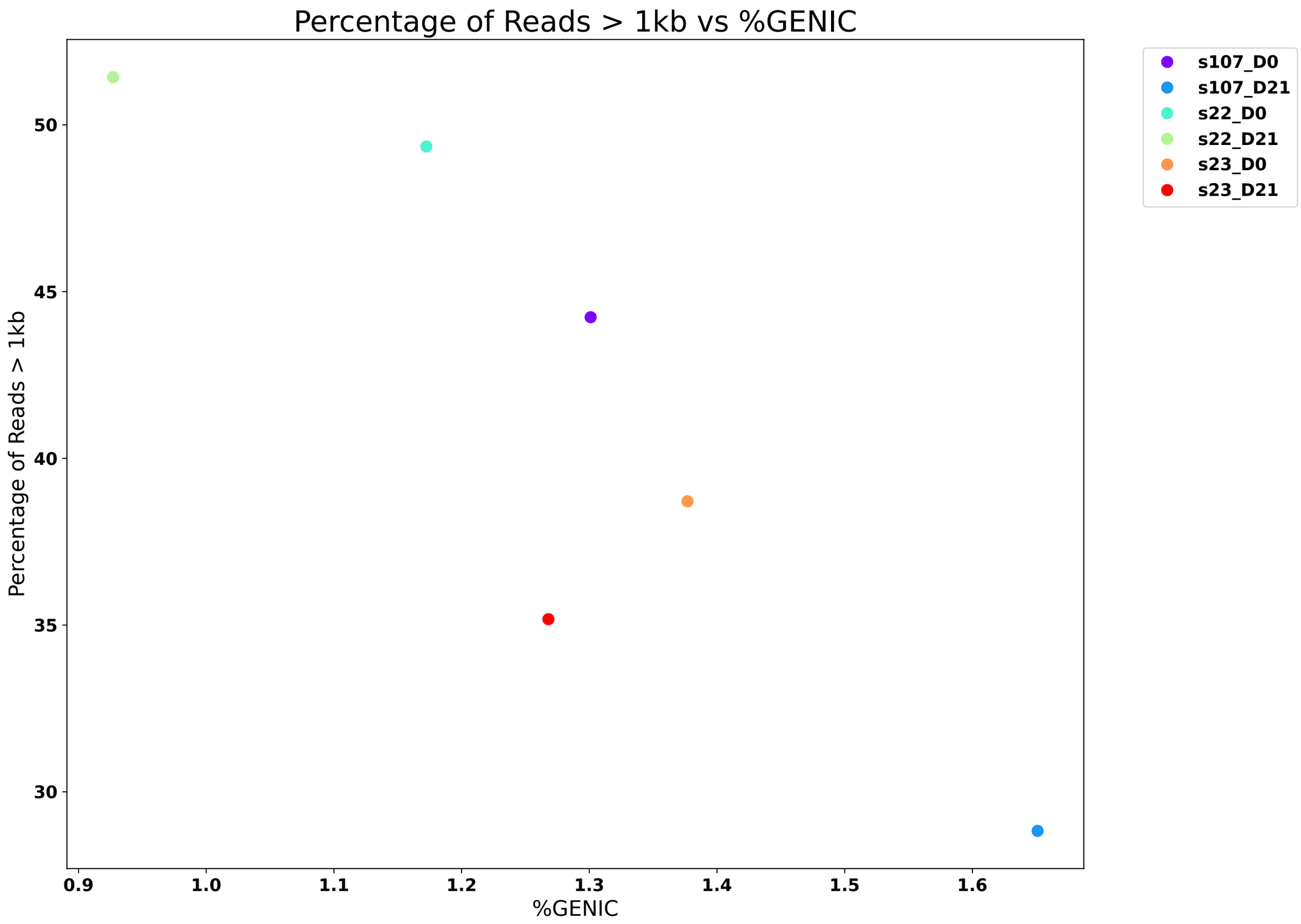

Percentage of Reads > 1kb vs %FSM
Percentage of Reads > 1kb
%FSM
50
45
40
35
30
45
50
55
60
65
70
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21

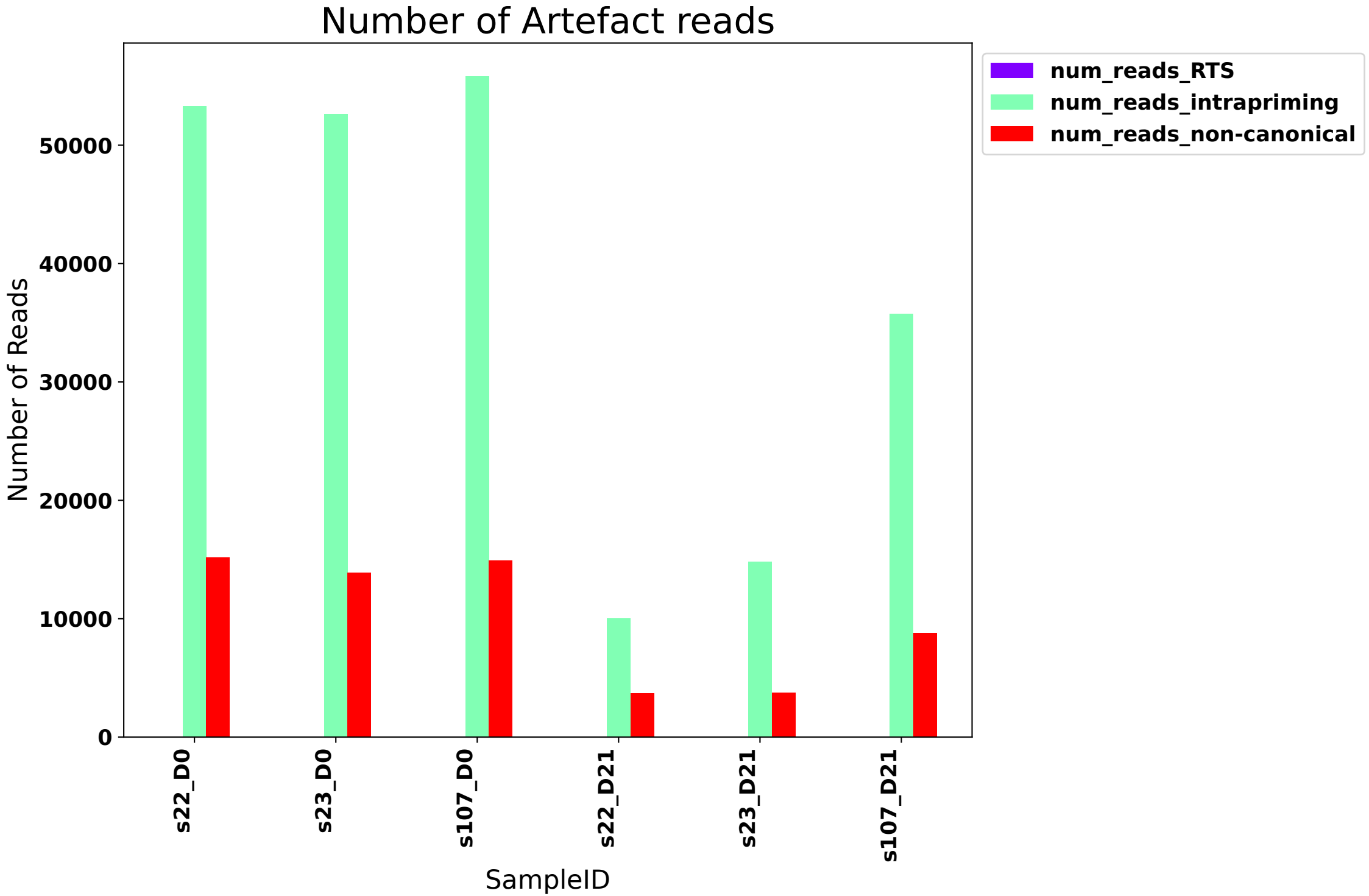

Percentage of Reads > 1kb vs %ISM
Percentage of Reads > 1kb
%ISM
50
45
40
35
30
20
25
30
35
40
45
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21

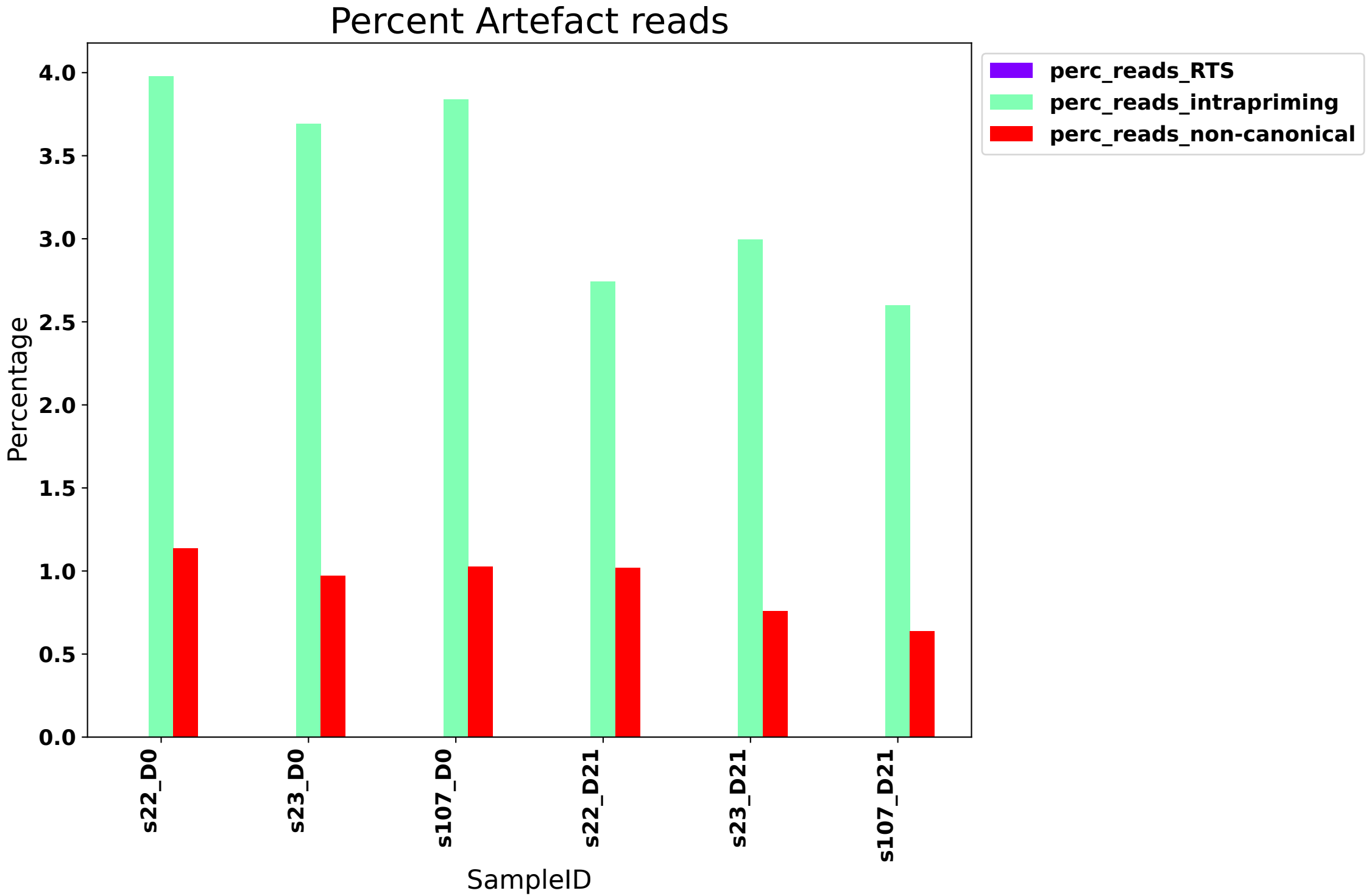

Percentage of Reads > 1kb vs %NIC
Percentage of Reads > 1kb
%NIC
50
45
40
35
30
1.2
1.4
1.6
1.8
2.0
2.2
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21

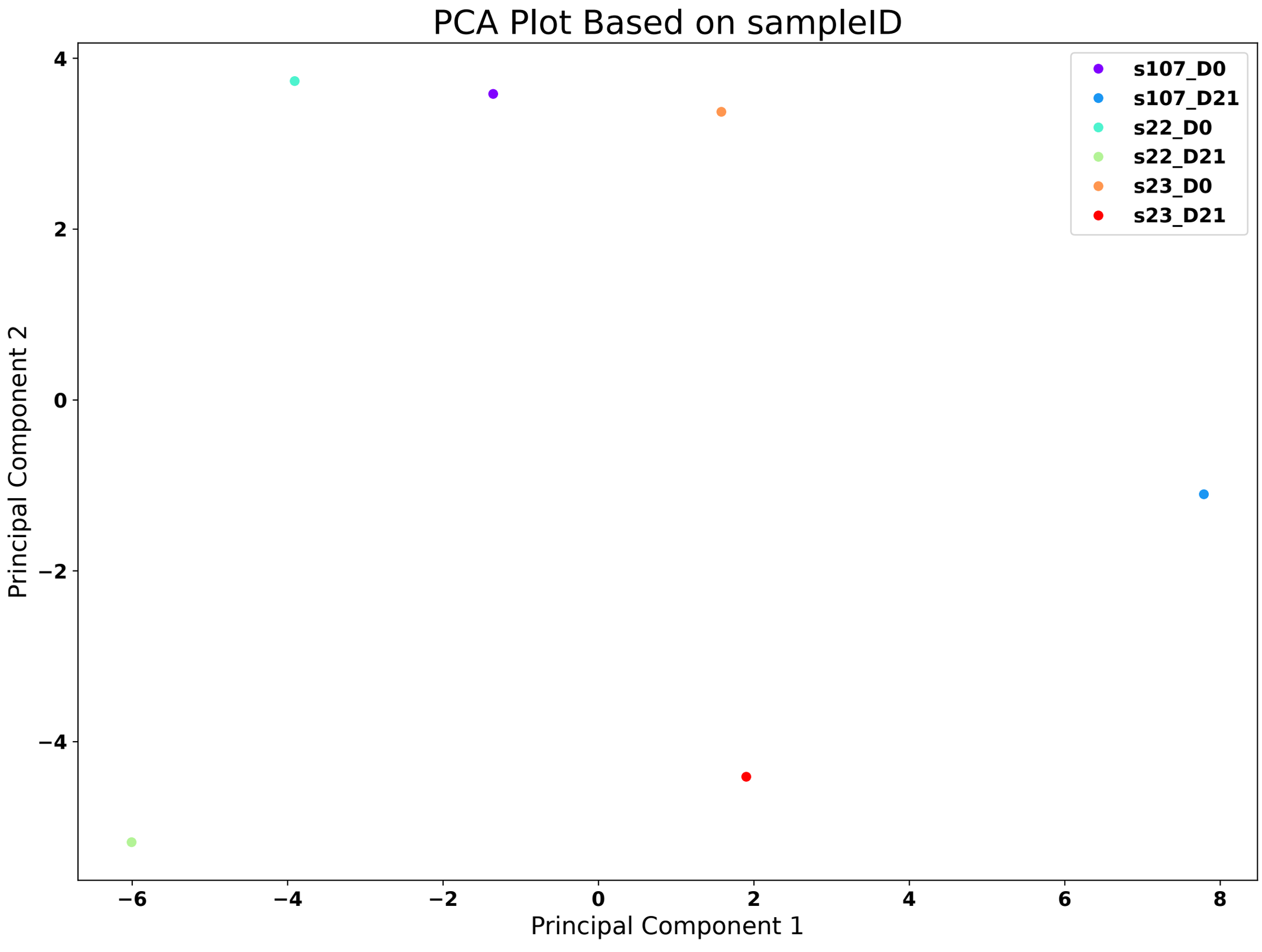

Percentage of Reads > 1kb vs %NNC
Percentage of Reads > 1kb
%NNC
50
45
40
35
30
3.5
4.0
4.5
5.0
5.5
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21

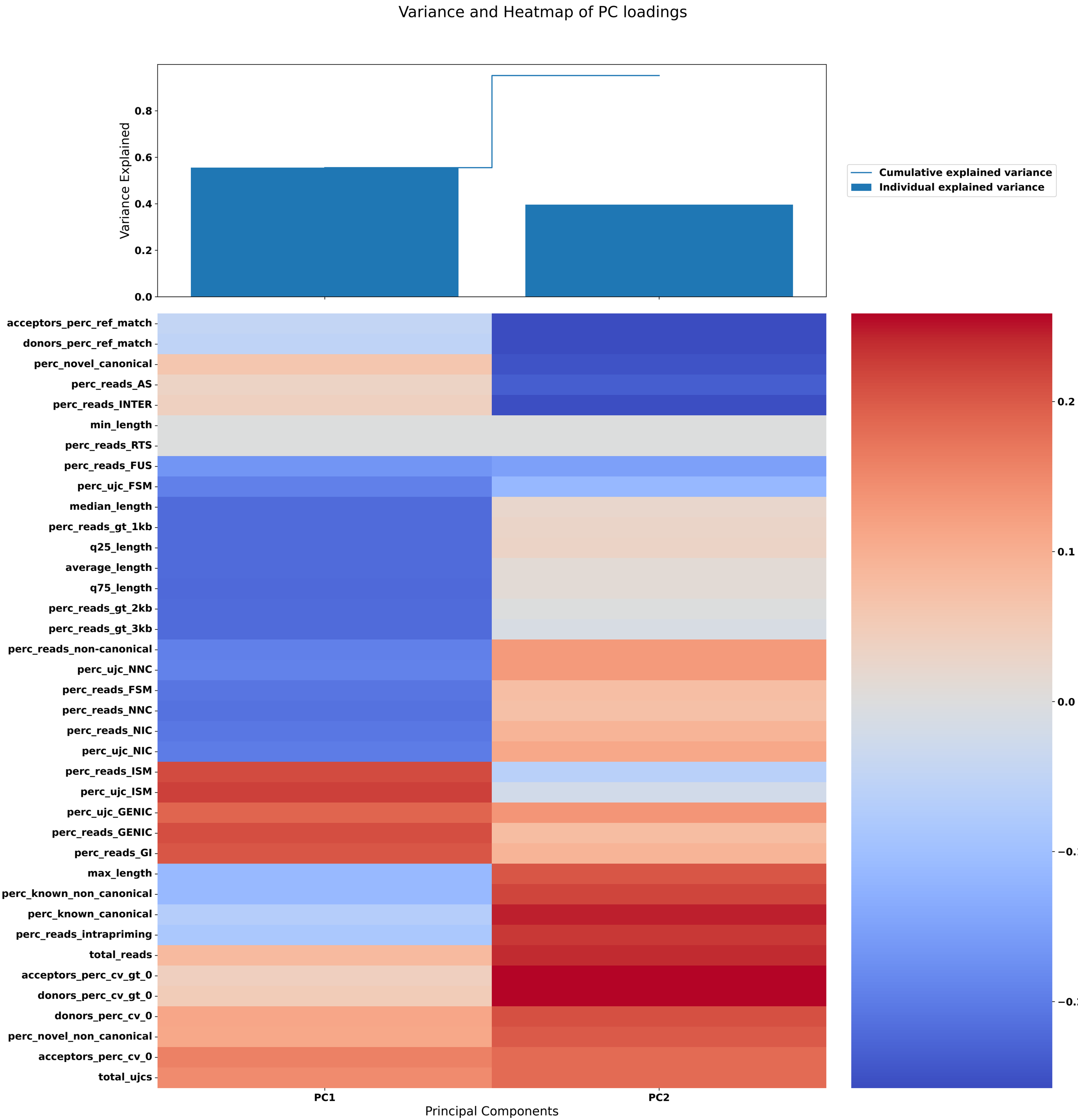

Percentage of Reads > 1kb vs %GI
Percentage of Reads > 1kb
%GI
50
45
40
35
30
0.6
0.8
1.0
1.2
1.4
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21

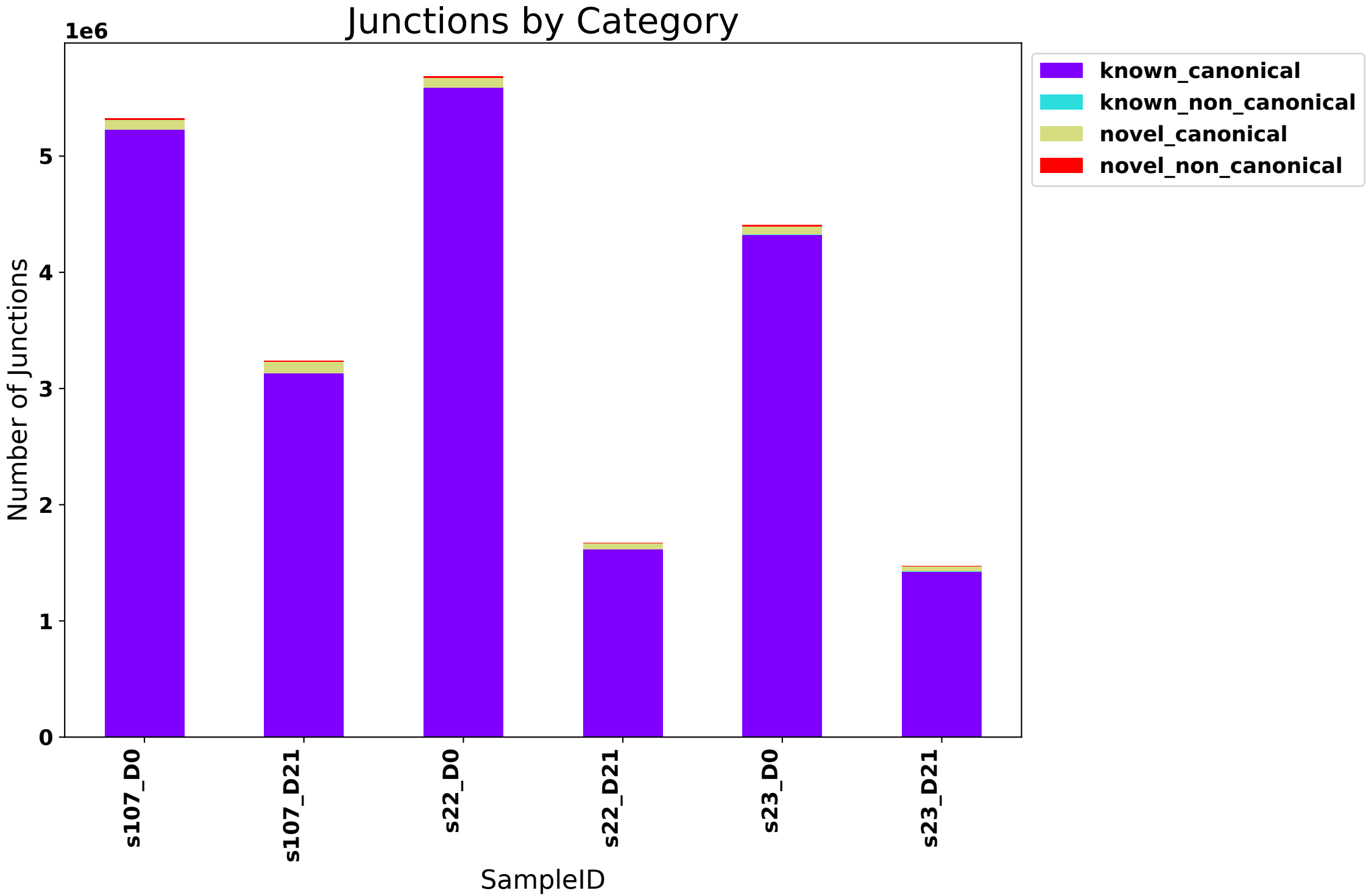

Percentage of Reads > 1kb vs %GENIC
Percentage of Reads > 1kb
%GENIC
50
45
40
35
30
0.9
1.0
1.1
1.2
1.3
1.4
1.5
1.6
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21

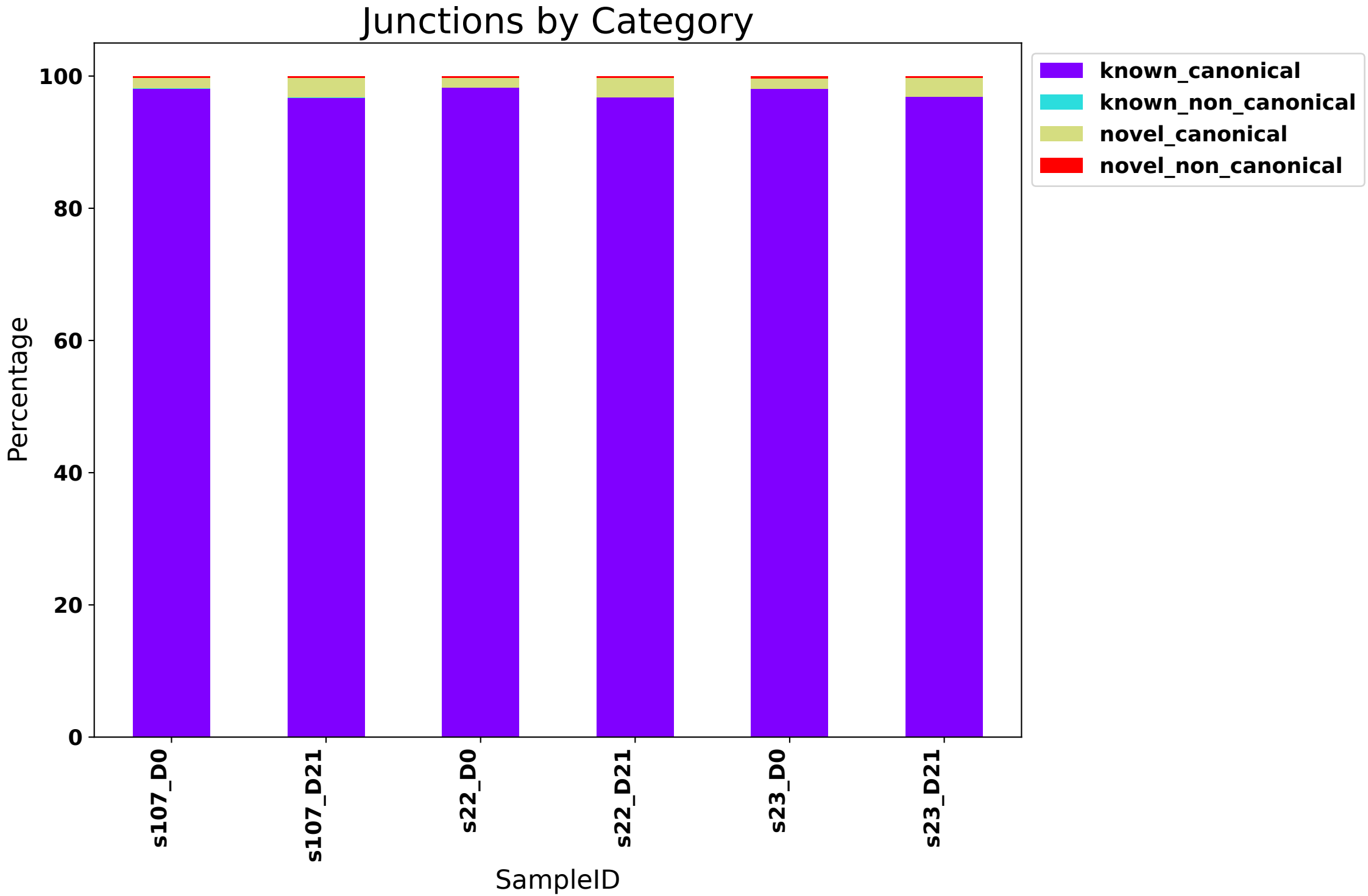

Number of Artefact reads
num_reads_RTS
num_reads_intrapriming
num_reads_non-canonical
Number of Reads
0
10000
20000
30000
40000
50000
s22_D0
s23_D0
s107_D0
s22_D21
s23_D21
s107_D21
SampleID

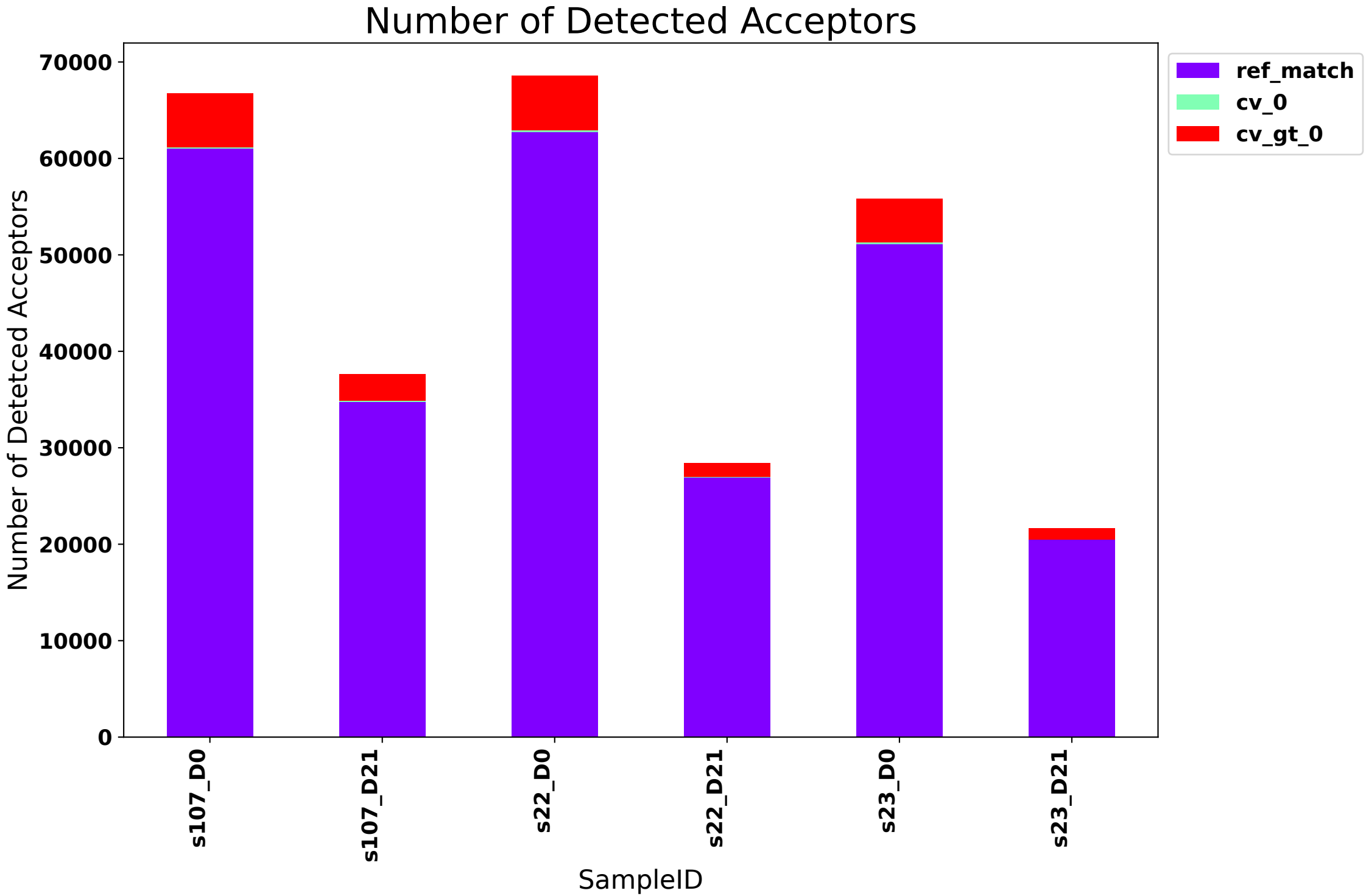

Percent Artefact reads
perc_reads_RTS
perc_reads_intrapriming
perc_reads_non-canonical
Percentage
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
s22_D0
s23_D0
s107_D0
s22_D21
s23_D21
s107_D21
SampleID

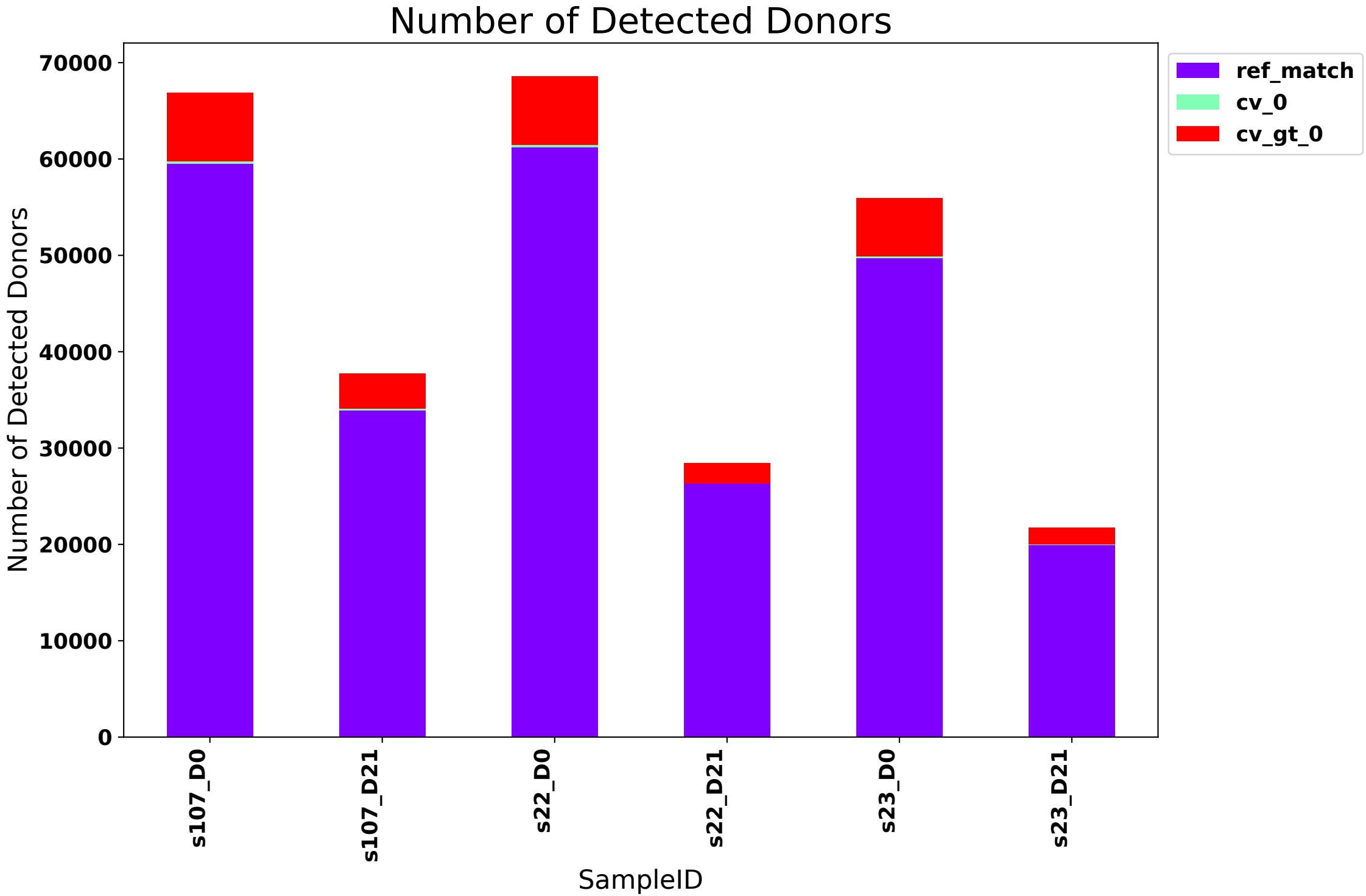

PCA Plot Based on sampleID
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
Principal Component 2
Principal Component 1
4
2
0
−2
−4
−6
−4
−2
0
2
4
6
8

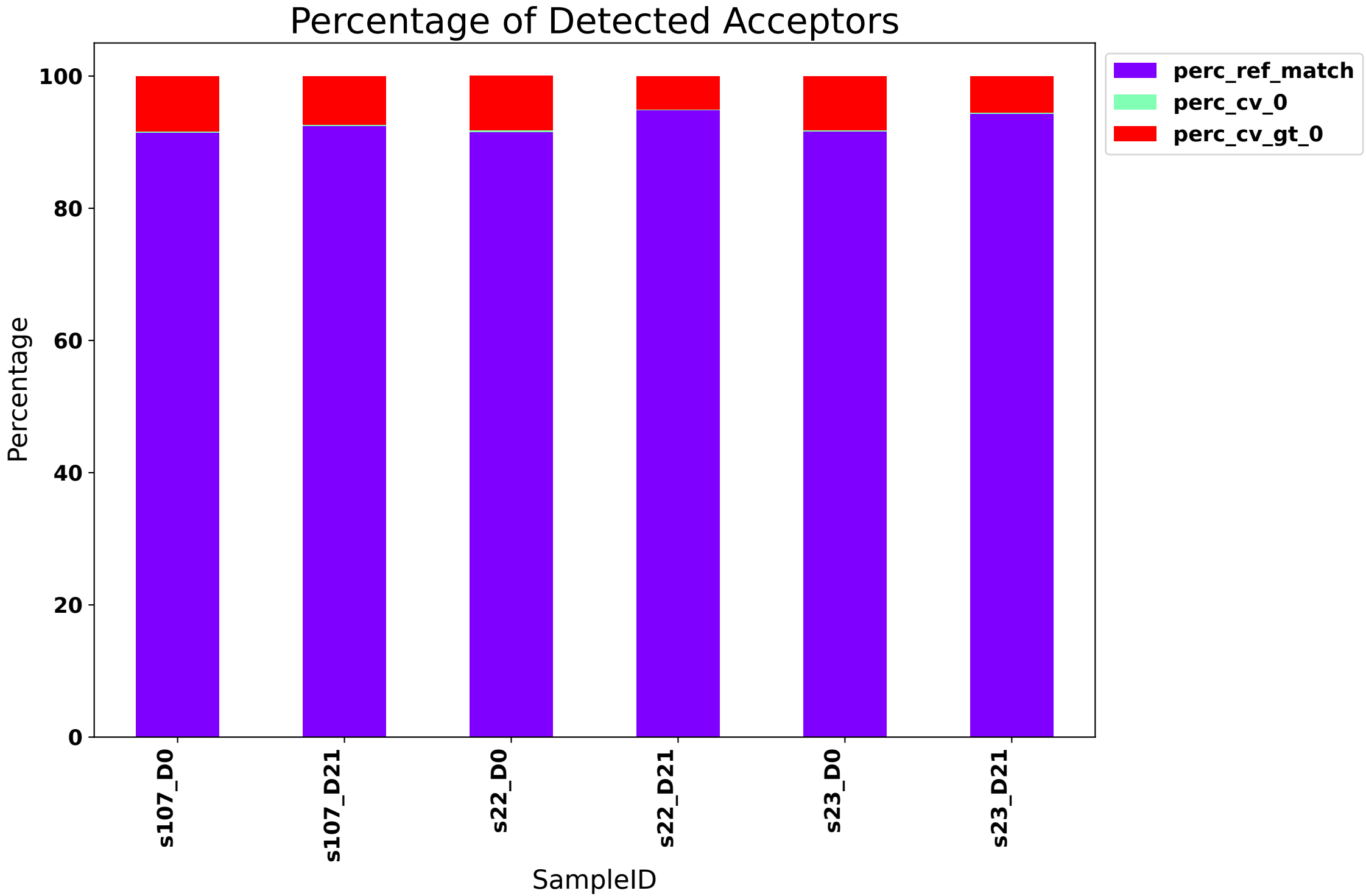

Variance and Heatmap of PC loadings
Variance Explained
0.0
0.2
0.4
0.6
0.8
Cumulative explained variance
Individual explained variance
acceptors_perc_ref_match
donors_perc_ref_match
perc_novel_canonical
perc_reads_AS
perc_reads_INTER
min_length
perc_reads_RTS
perc_reads_FUS
perc_ujc_FSM
median_length
perc_reads_gt_1kb
q25_length
average_length
q75_length
perc_reads_gt_2kb
perc_reads_gt_3kb
perc_reads_non-canonical
perc_ujc_NNC
perc_reads_FSM
perc_reads_NNC
perc_reads_NIC
perc_ujc_NIC
perc_reads_ISM
perc_ujc_ISM
perc_ujc_GENIC
perc_reads_GENIC
perc_reads_GI
max_length
perc_known_non_canonical
perc_known_canonical
perc_reads_intrapriming
total_reads
acceptors_perc_cv_gt_0
donors_perc_cv_gt_0
donors_perc_cv_0
perc_novel_non_canonical
acceptors_perc_cv_0
total_ujcs
PC1
PC2
Principal Components
0.2
0.1
0.0
−0.1
−0.2

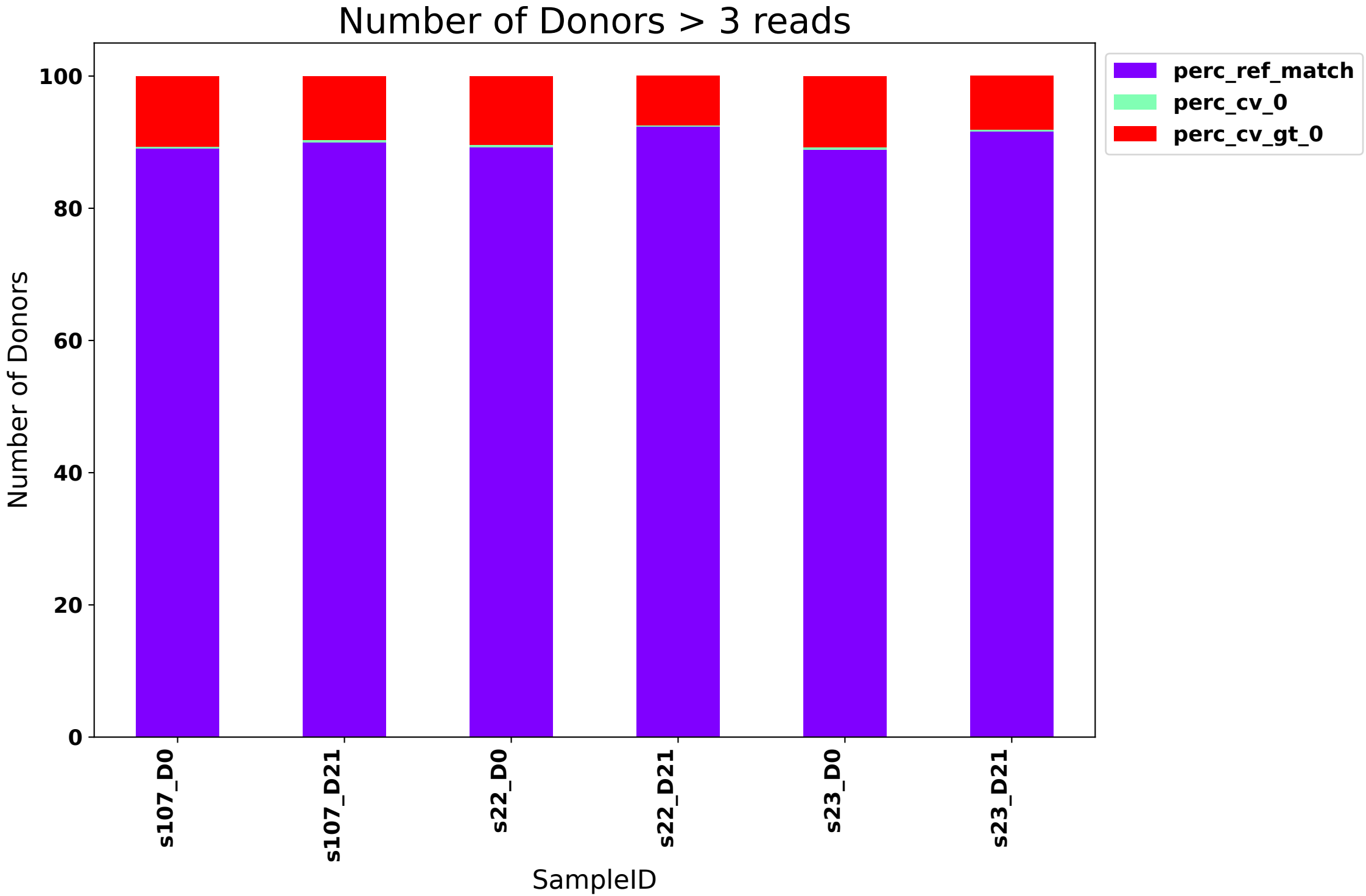

Junctions by Category
1e6
Number of Junctions
0
1
2
3
4
5
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID
known_canonical
known_non_canonical
novel_canonical
novel_non_canonical


Junctions by Category
Percentage
100
80
60
40
20
0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID
known_canonical
known_non_canonical
novel_canonical
novel_non_canonical


Number of Detected Acceptors
Number of Detetced Acceptors
70000
60000
50000
40000
30000
20000
10000
0
ref_match
cv_0
cv_gt_0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID


Number of Detected Donors
Number of Detected Donors
70000
60000
50000
40000
30000
20000
10000
0
ref_match
cv_0
cv_gt_0
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
SampleID


Percentage of Detected Acceptors
Percentage
SampleID
0
20
40
60
80
100
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
perc_ref_match
perc_cv_0
perc_cv_gt_0


Number of Donors > 3 reads
Number of Donors
SampleID
0
20
40
60
80
100
s107_D0
s107_D21
s22_D0
s22_D21
s23_D0
s23_D21
perc_ref_match
perc_cv_0
perc_cv_gt_0